

ALADIN EEG Validation Suite – S03-DMT

30 Research-Grade Plots + Summary

Author: Bucurenciu Mihai Alexandru (Aladin)
Email: aladinibz@proton.me

Summary Statistics

ALADIN S03-DMT Validation Summary

=====

PAC (Tort MI): 0.0008

$\Delta\alpha$ pre $t \approx 69.789062$: 4.9197

$\Delta\alpha$ post $t \approx 69.789062$: 4.2508

$\Delta\alpha$ drop %: 13.6%

Permutation p $\Delta\alpha$: 0.3718

Permutation p Gamma: 0.8914

Cohen's d $\Delta\alpha$: 0.298

Cohen's d Gamma: -0.178

Granger raw p-values: [np.float64(6.226375464314537e-53), np.float64(4.8010283586e-53)]

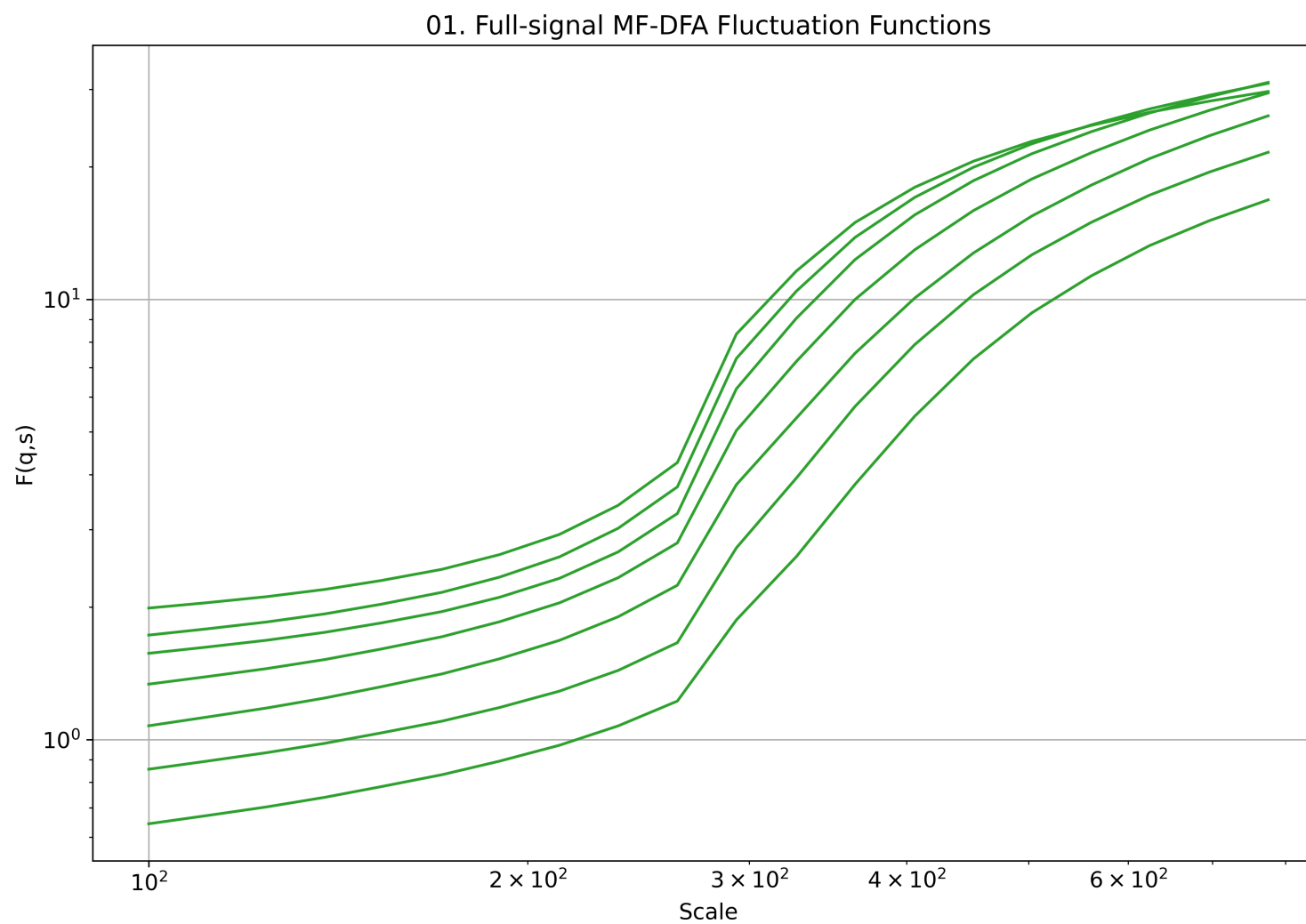
Granger FDR corrected p-values: [6.226375464314537e-53, 6.001285448308643e-84]

$\Delta\alpha$ at t_{peak} : 5.9073

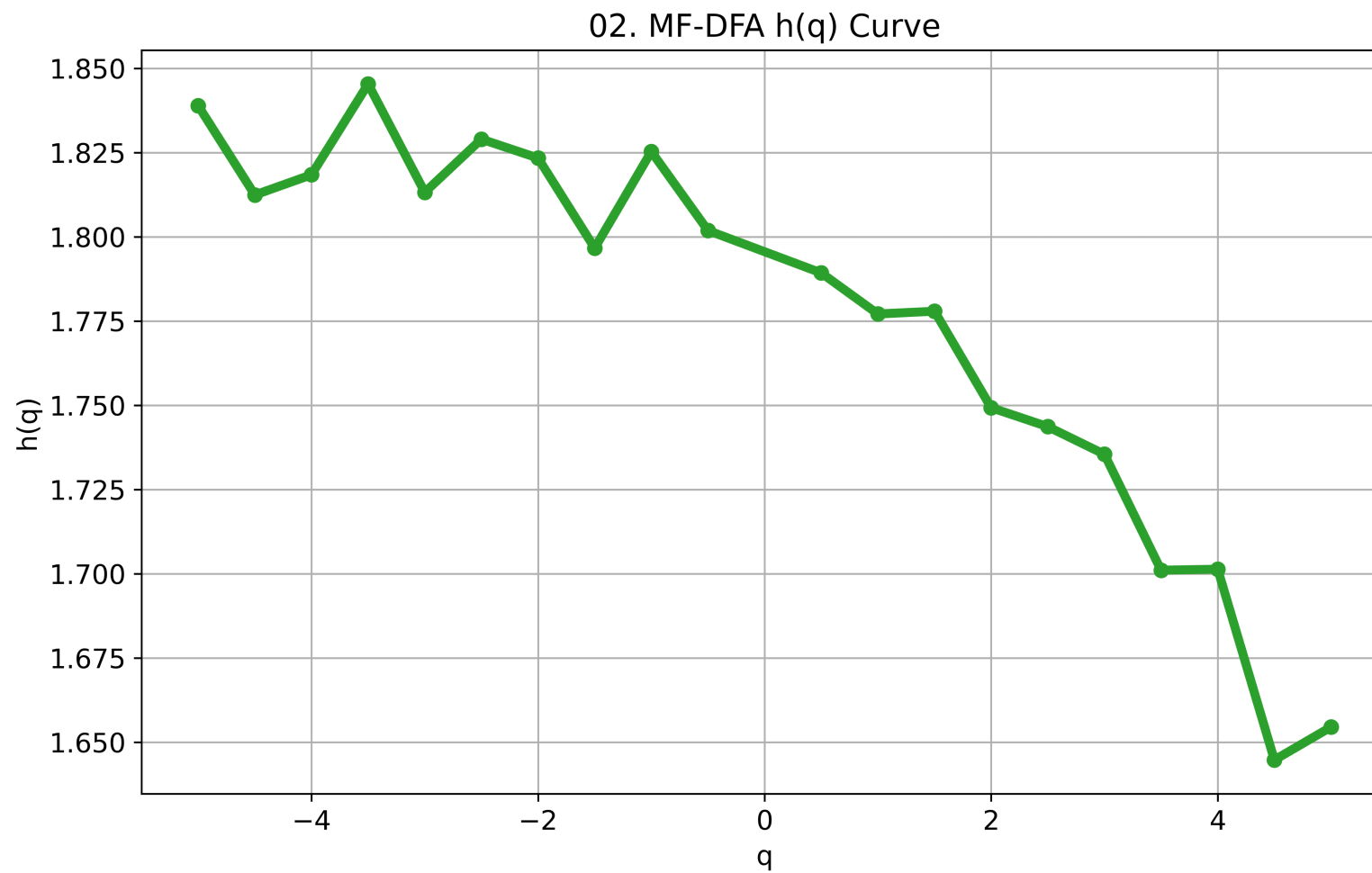
$\Delta\alpha$ min: 1.2349

$\Delta\alpha$ max: 11.3586

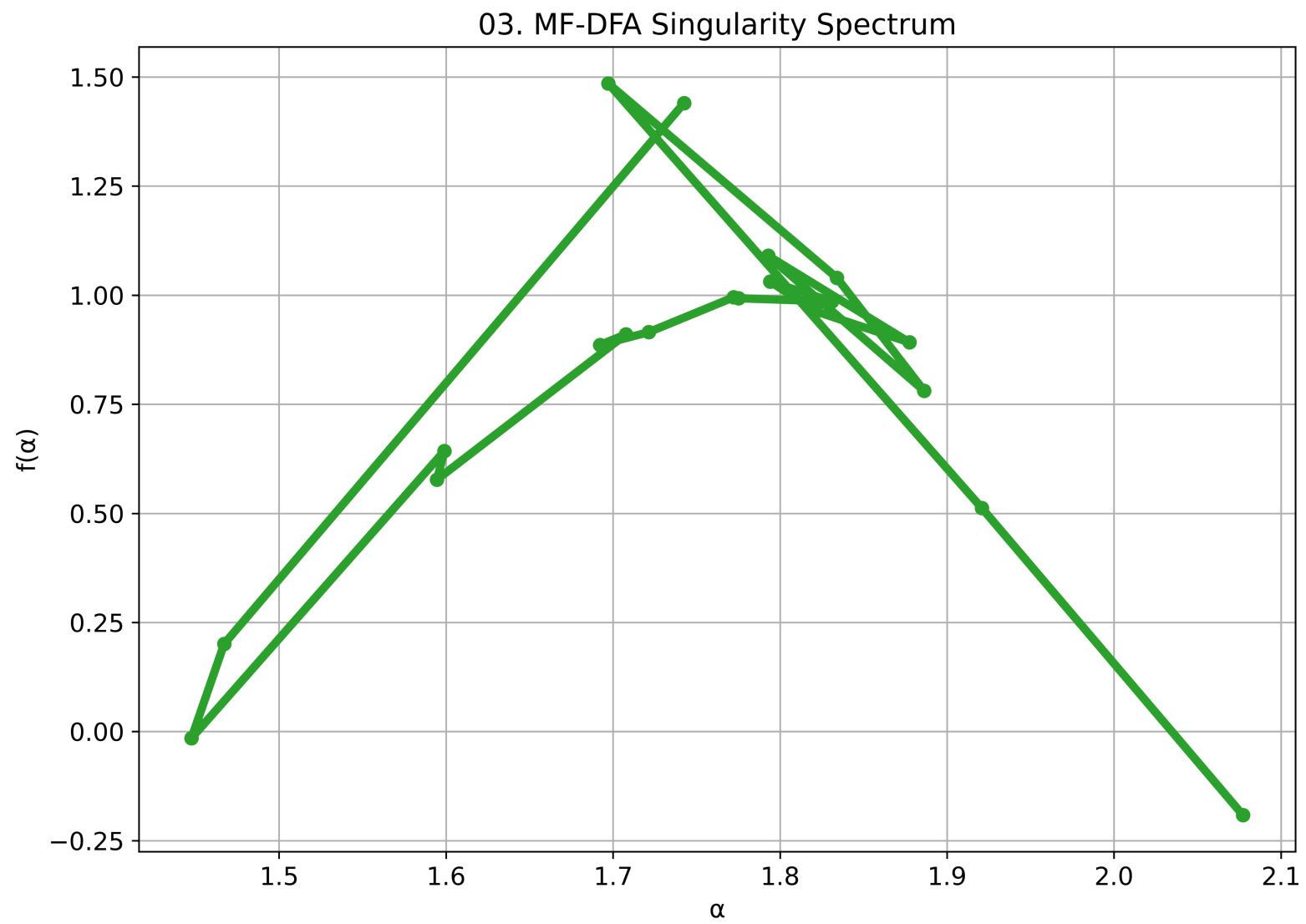
Z-score at t_{peak} : 0.43



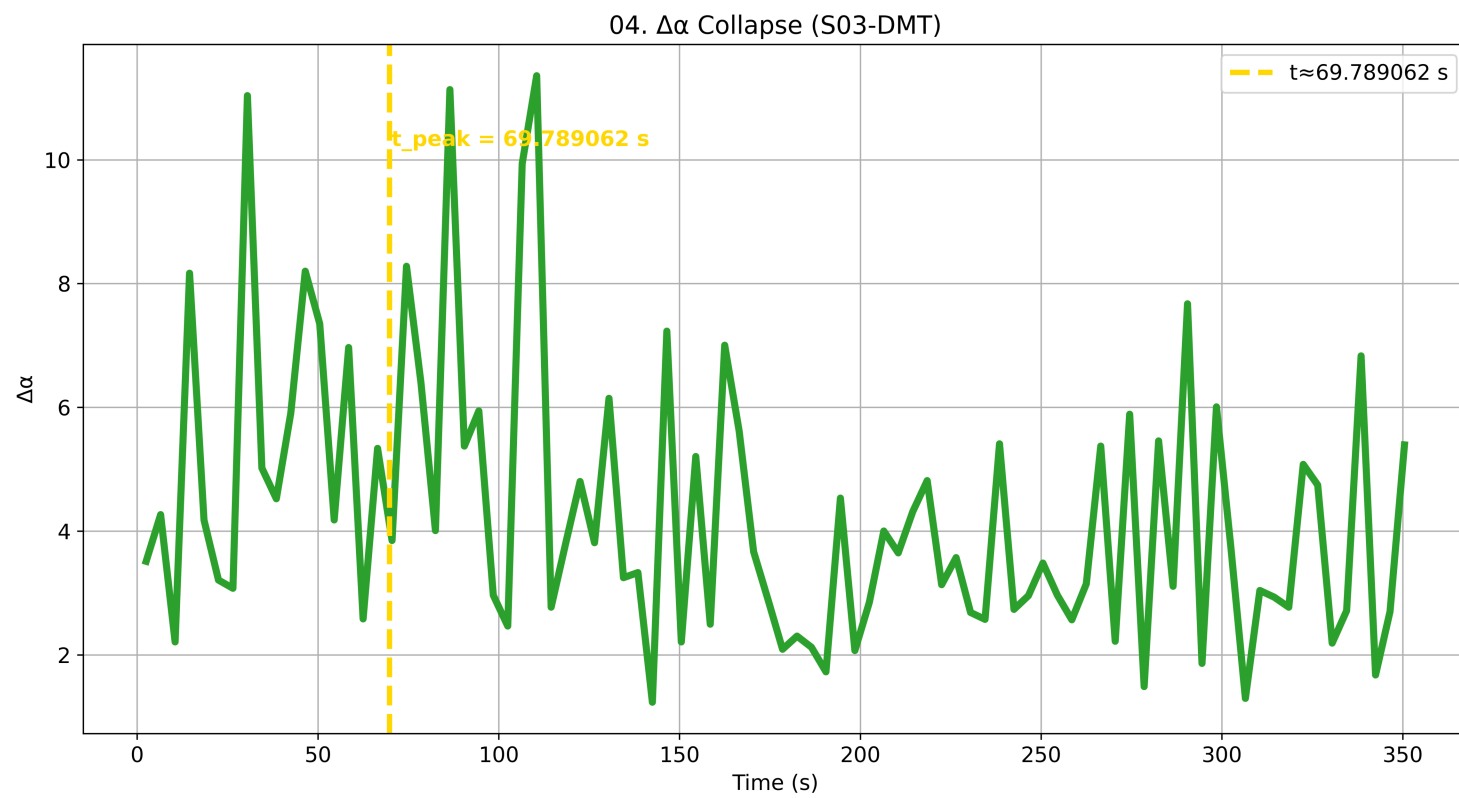
Full-signal MF-DFA fluctuation functions across q orders ($t_{\text{peak}} \approx 69.789062$ s)



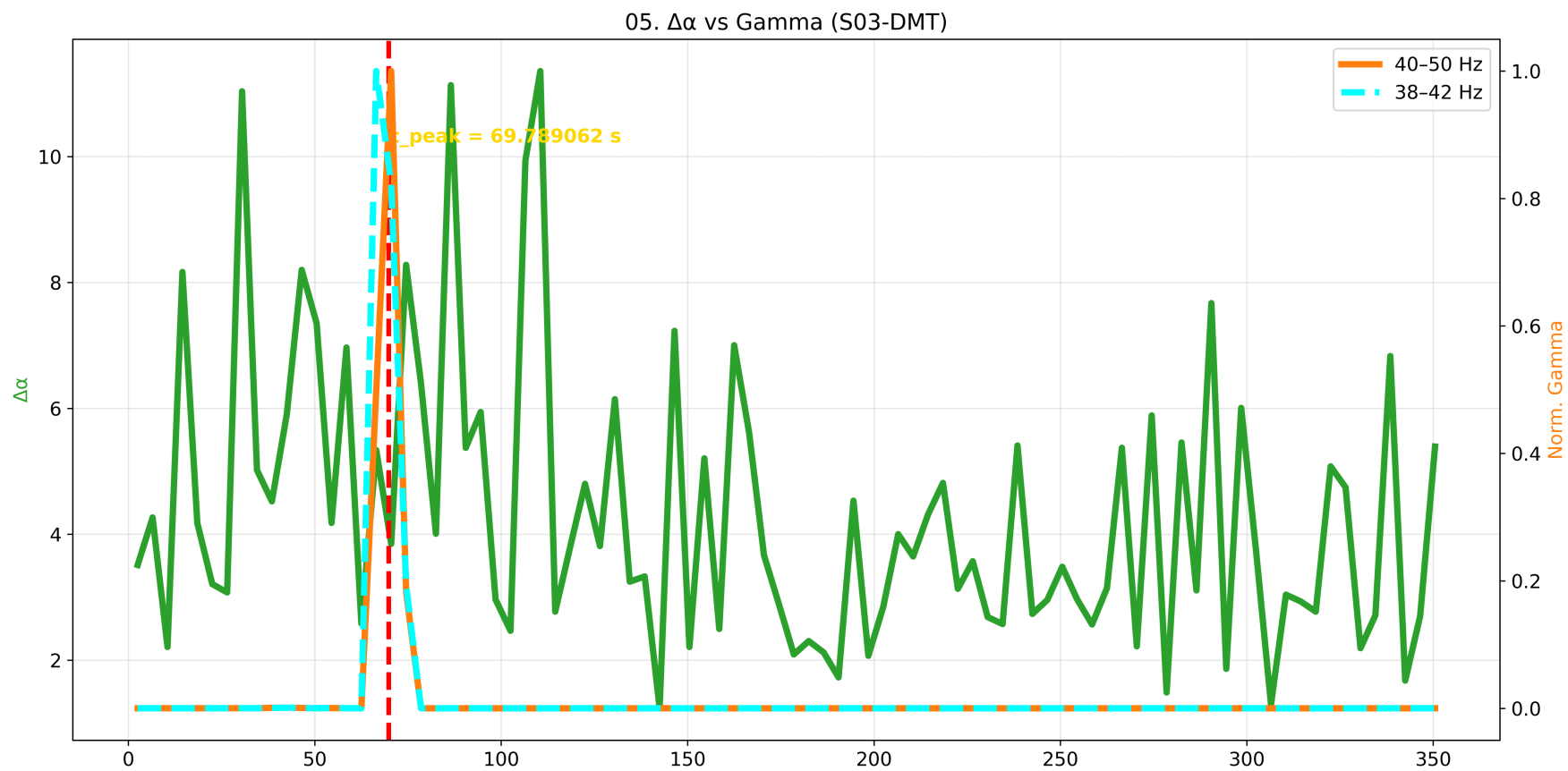
MF-DFA generalized Hurst exponent $h(q)$ curve showing multifractality ($t_{\text{peak}} \approx 69.789062$ s)



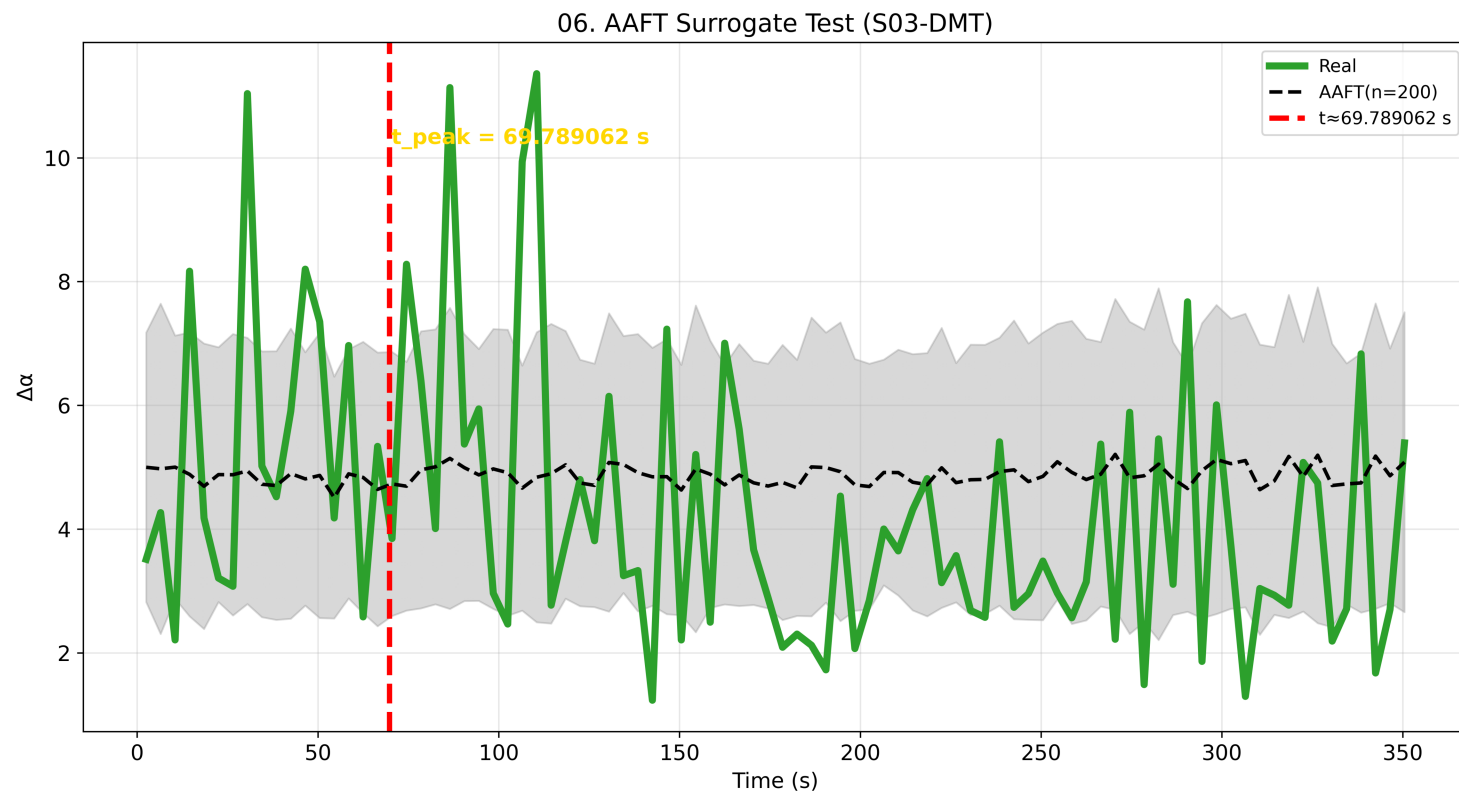
MF-DFA singularity spectrum $f(\alpha)$ – width indicates multifractal strength ($t_{\text{peak}} \approx 69.789062$ s)



Time-resolved $\Delta\alpha$ collapse with objective peak at 69.789062 s

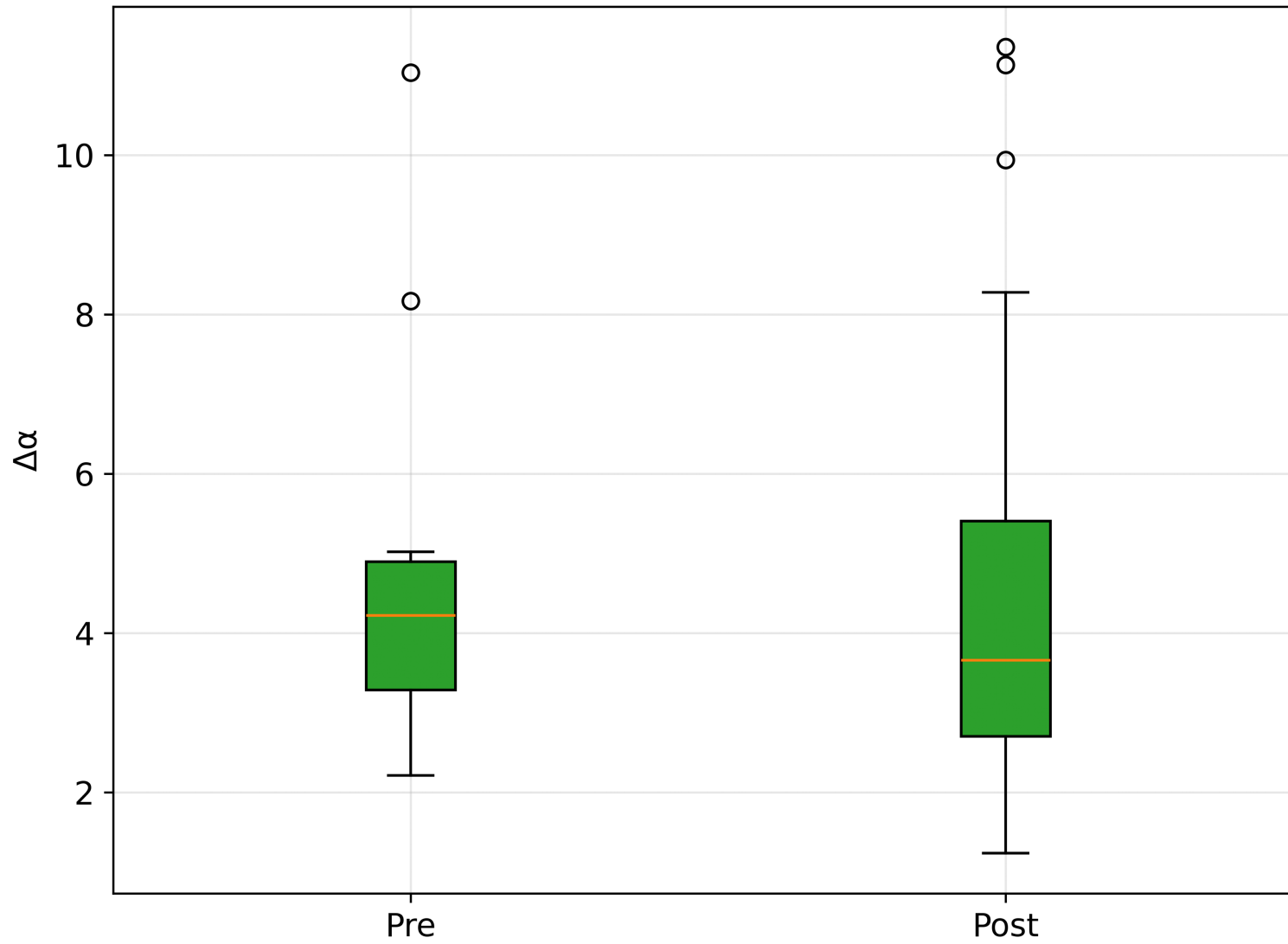


$\Delta\alpha$ collapse aligned with 40-50 Hz gamma power (peak at 69.789062 s)



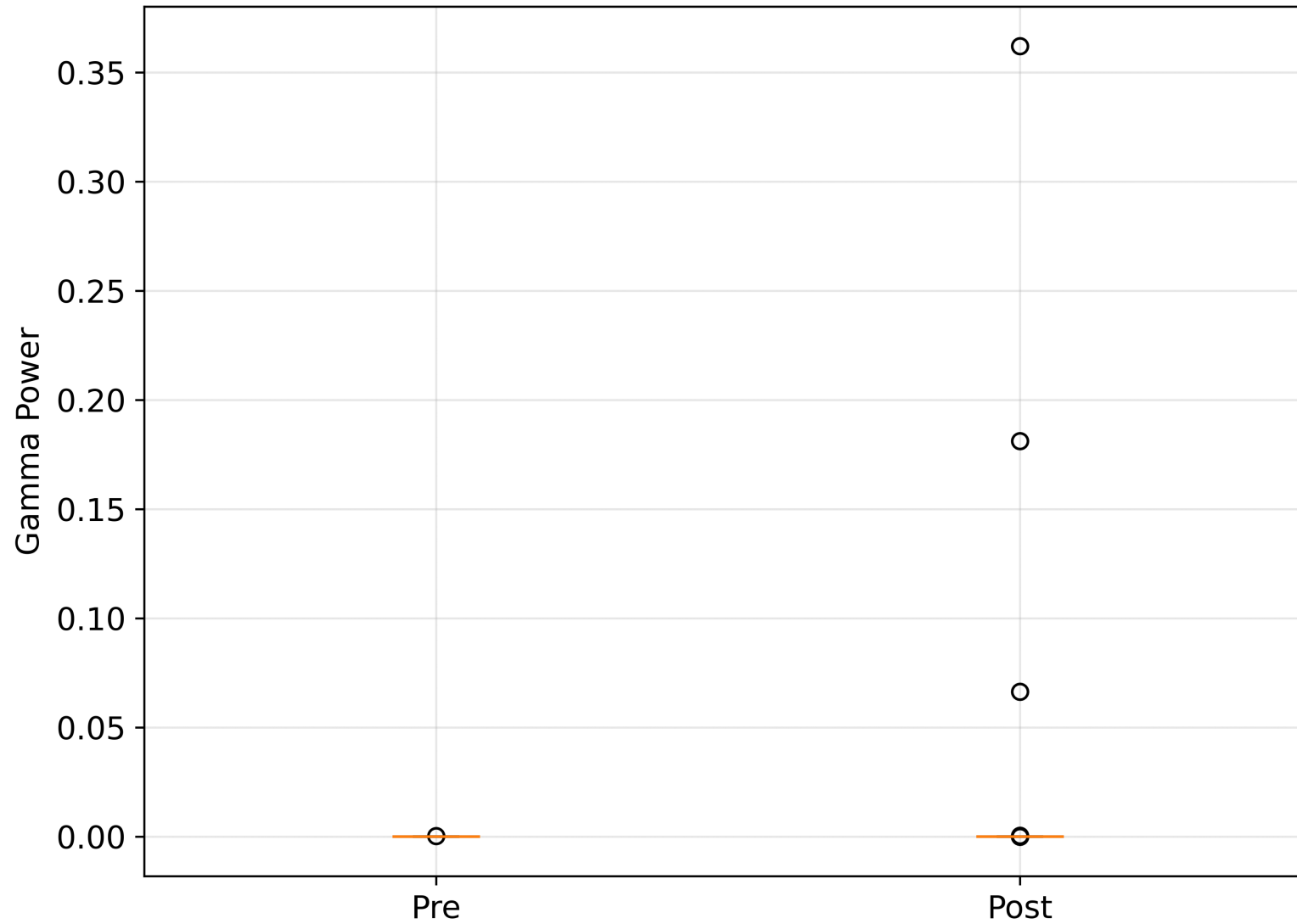
AAFT surrogate test with objective peak at 69.789062 s

07. $\Delta\alpha$ Pre/Post $t\approx 69.789062$ s (Perm $p=0.3718$)

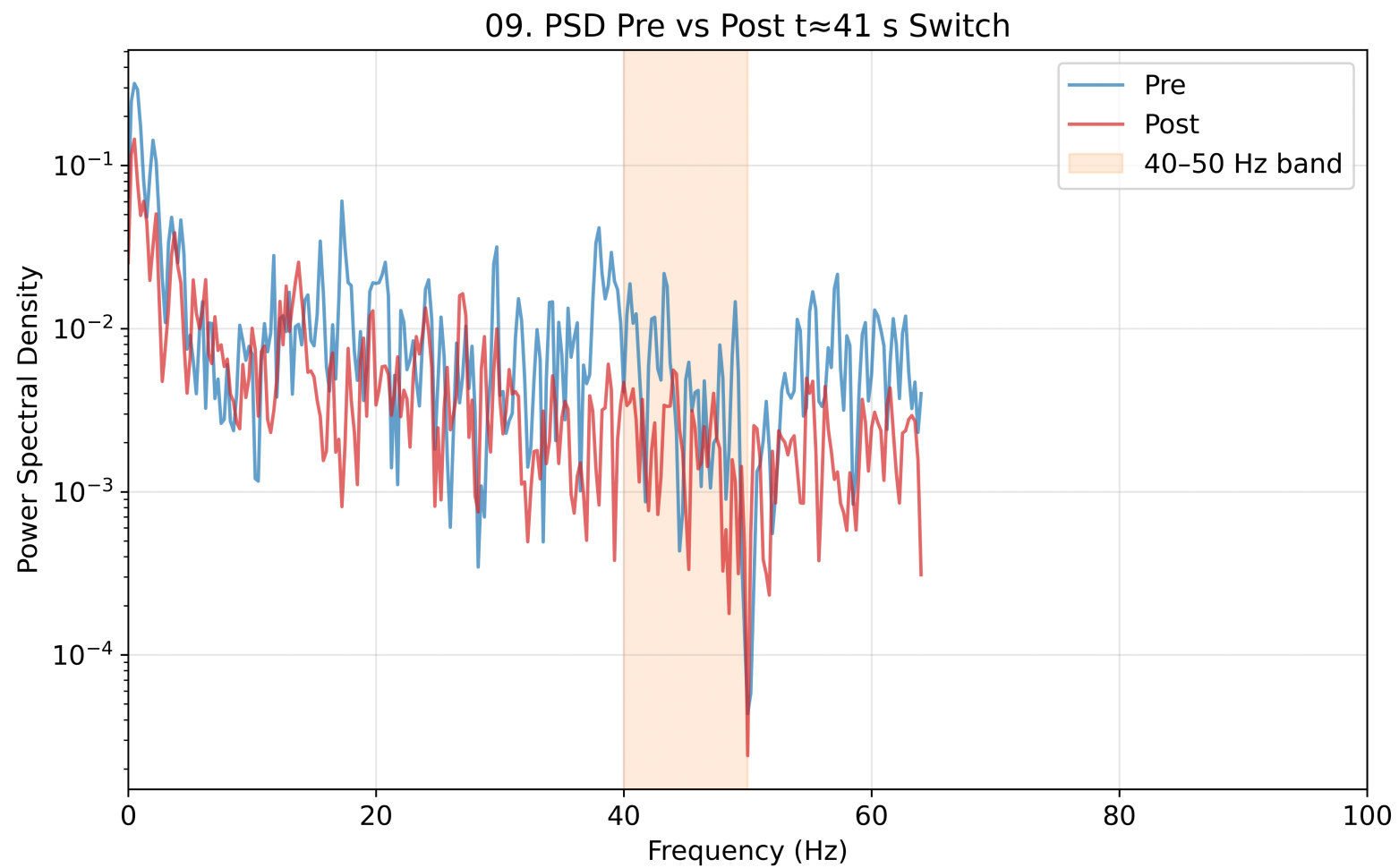


$\Delta\alpha$ pre vs post $t\approx 69.789062$ s with permutation test

08. Gamma 40-50 Hz Pre/Post $t \approx 69.789062$ s (Perm $p=0.8914$)

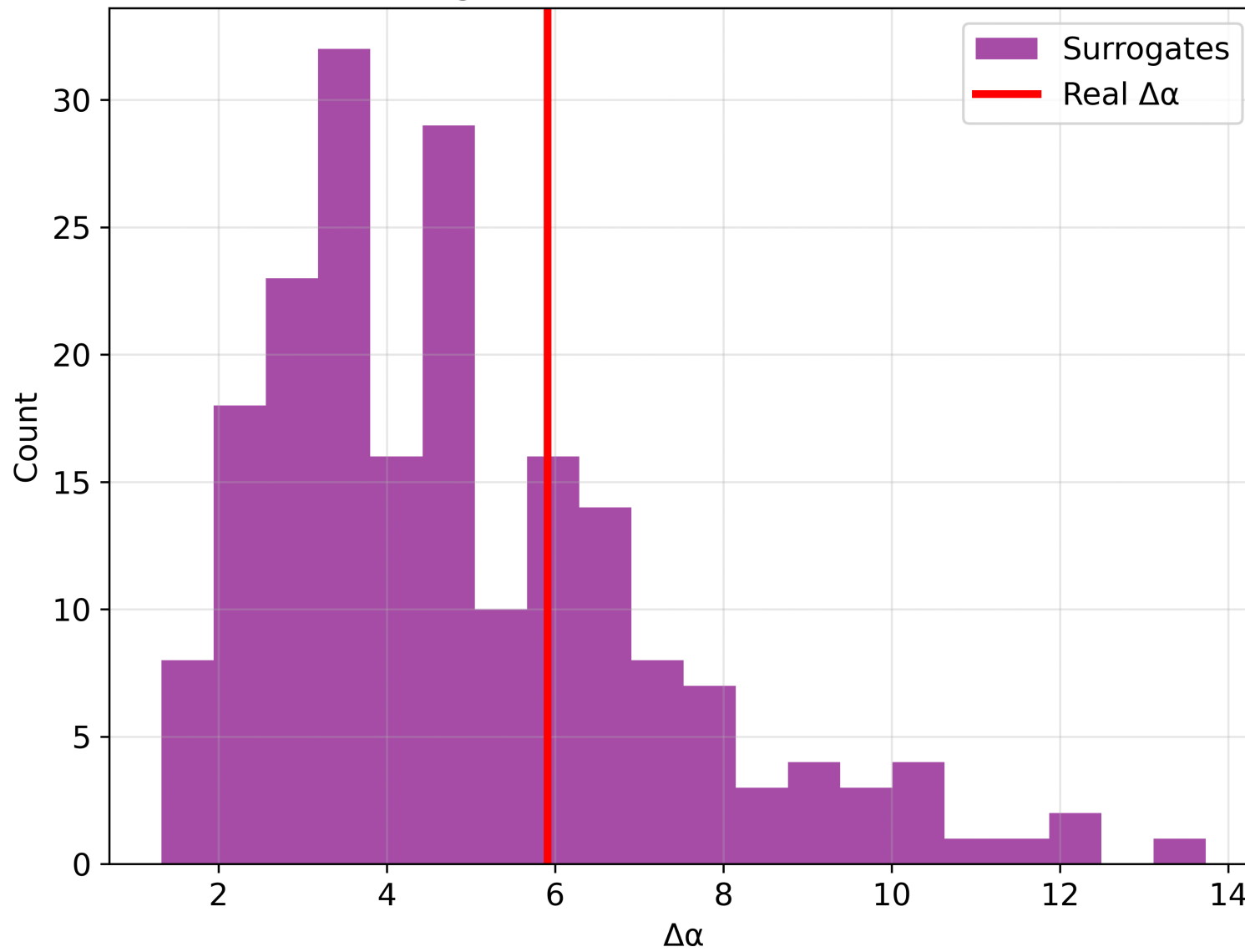


Gamma 40-50 Hz power pre vs post $t \approx 69.789062$ s with permutation test



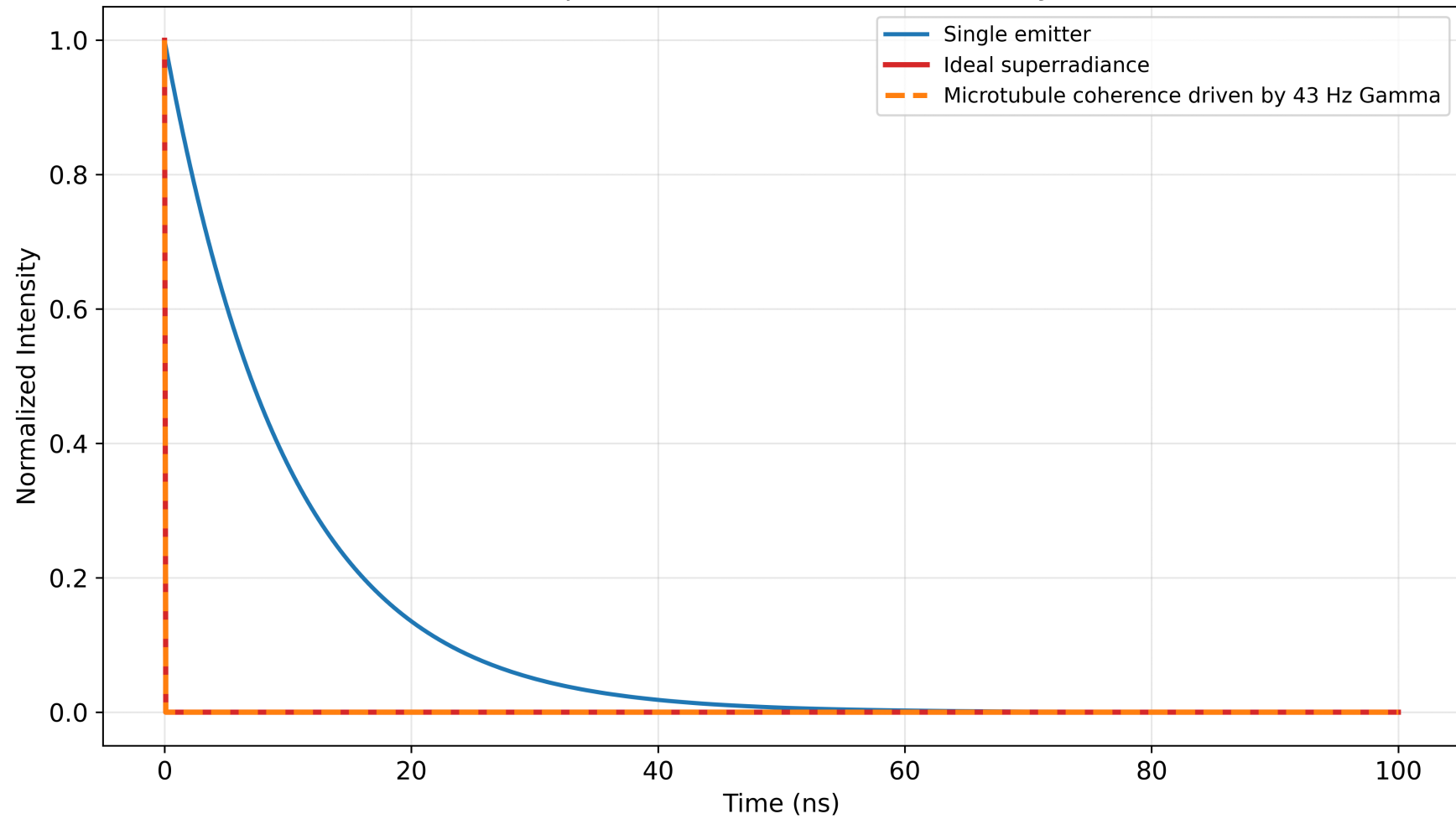
Power spectral density pre vs post $t \approx 69.789062$ s switch

10. Surrogate Z-score Distribution at $t \approx 41$ s

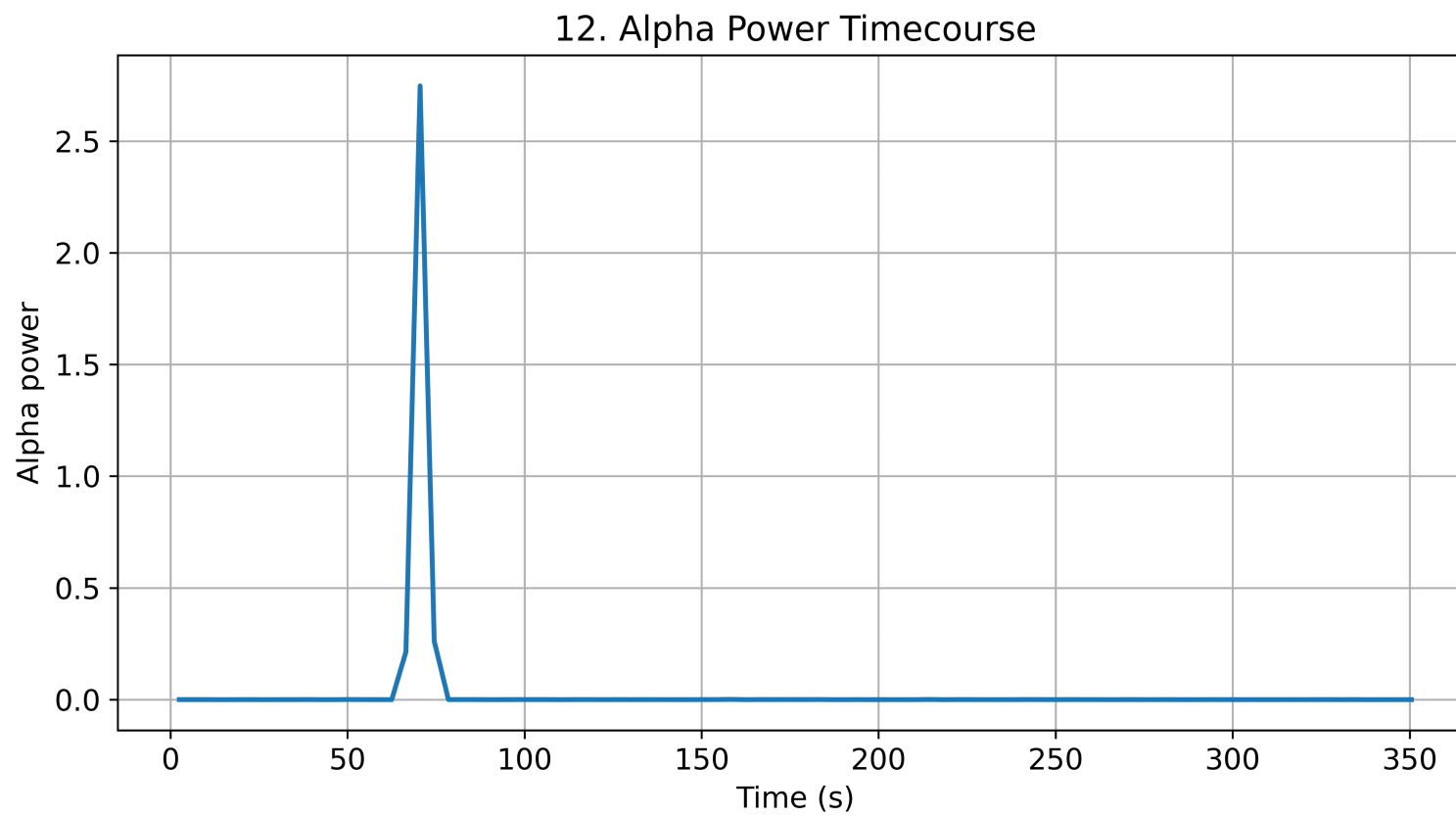


Surrogate distribution of $\Delta\alpha$ at peak time (real value marked, $t_{\text{peak}} \approx 69.789062$ s)

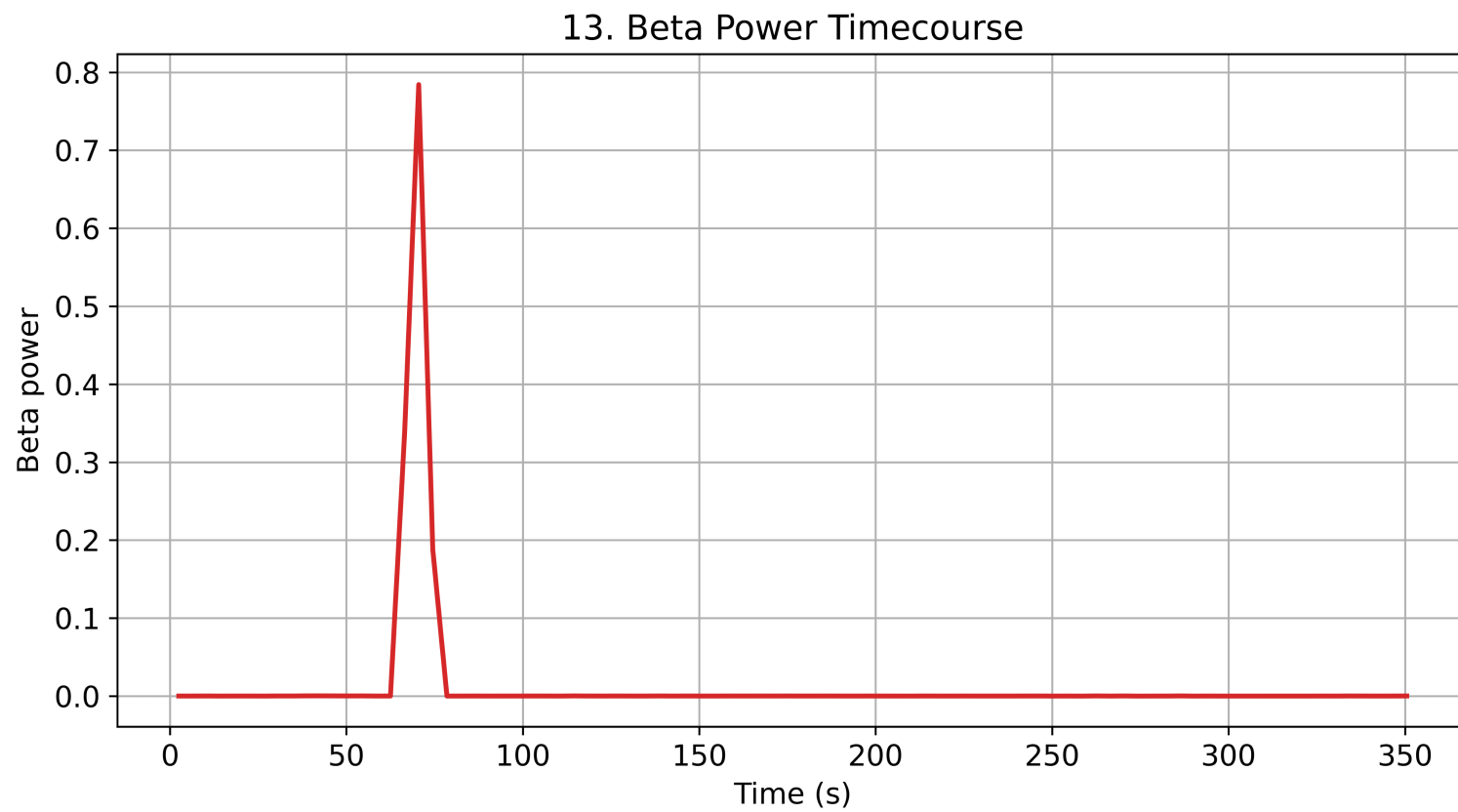
11. Microtubule Superradiance Coherence Driven by 43 Hz Gamma



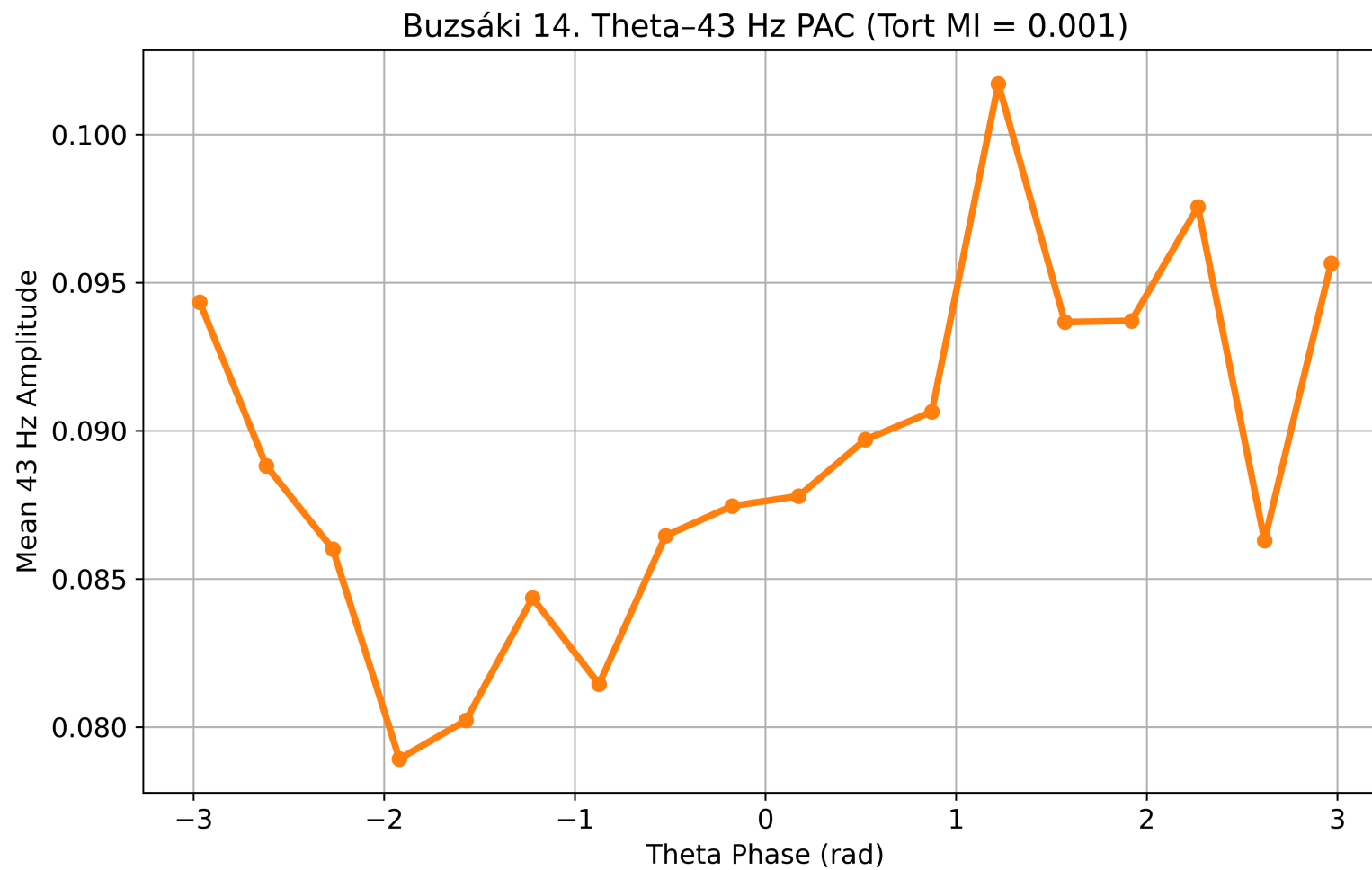
Microtubule Superradiance Coherence Driven by 43 Hz Gamma ($t_{\text{peak}} \approx 69.789062$ s)



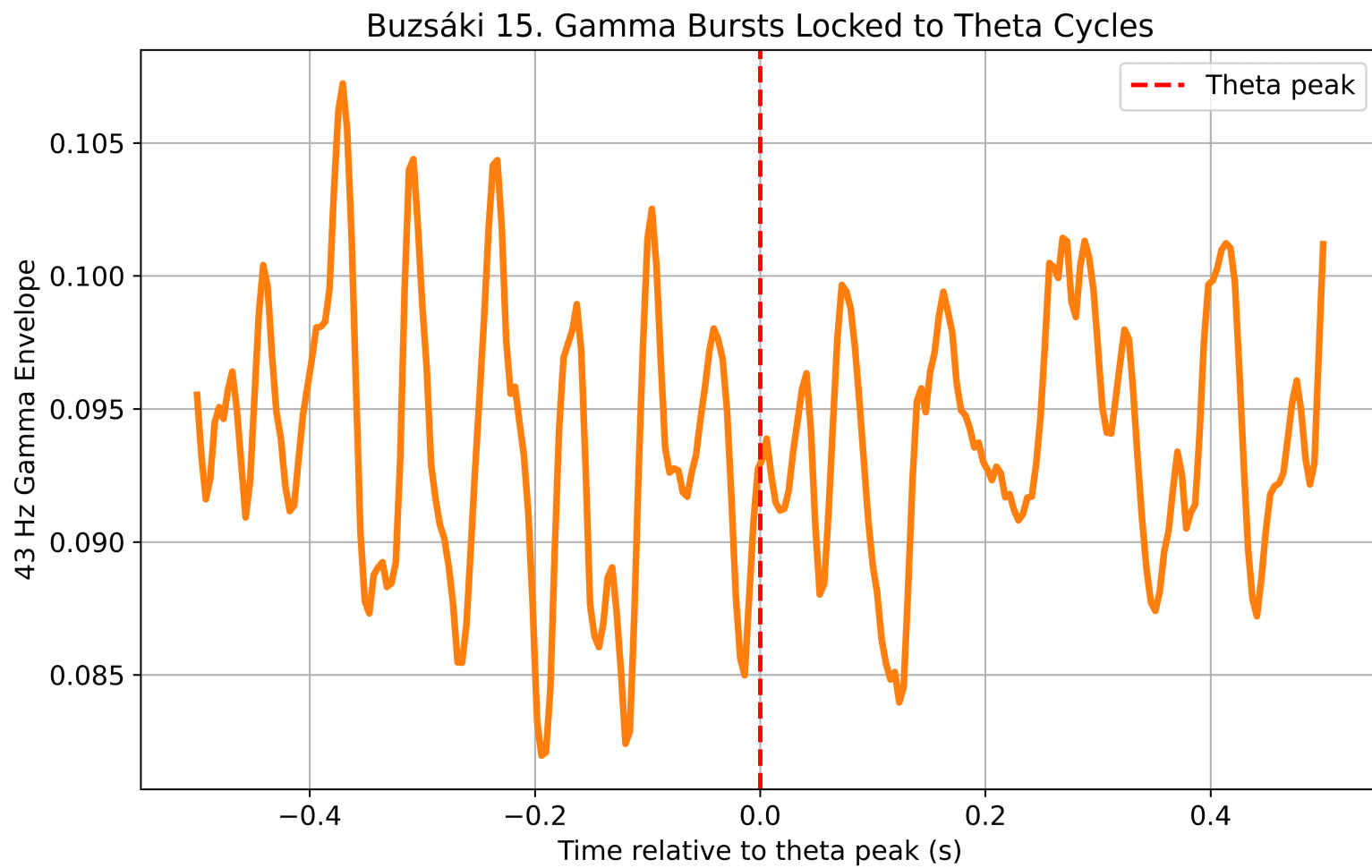
Alpha (8-12 Hz) power timecourse ($t_{\text{peak}} \approx 69.789062$ s)



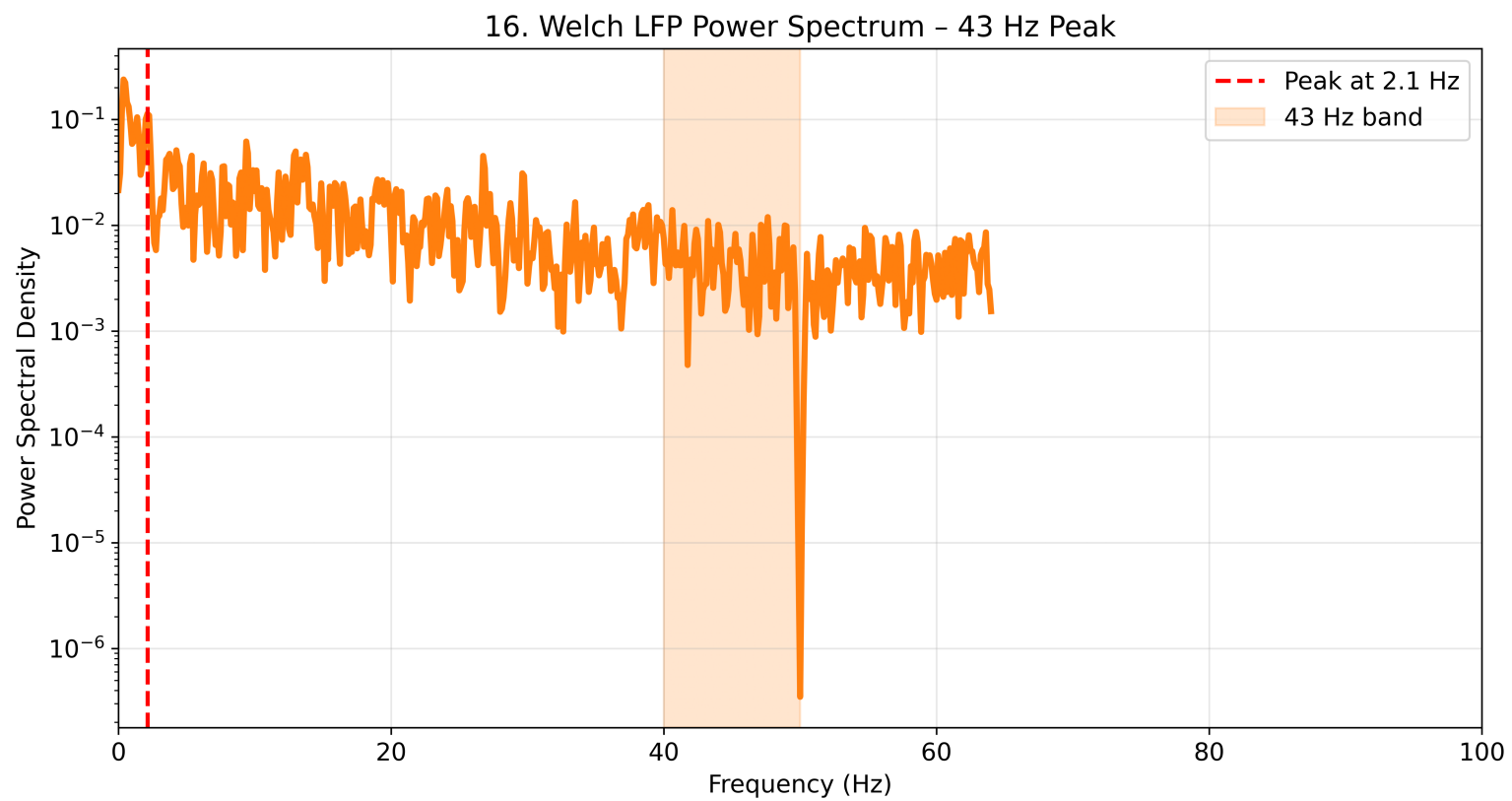
Beta (12-30 Hz) power timecourse ($t_{\text{peak}} \approx 69.789062$ s)



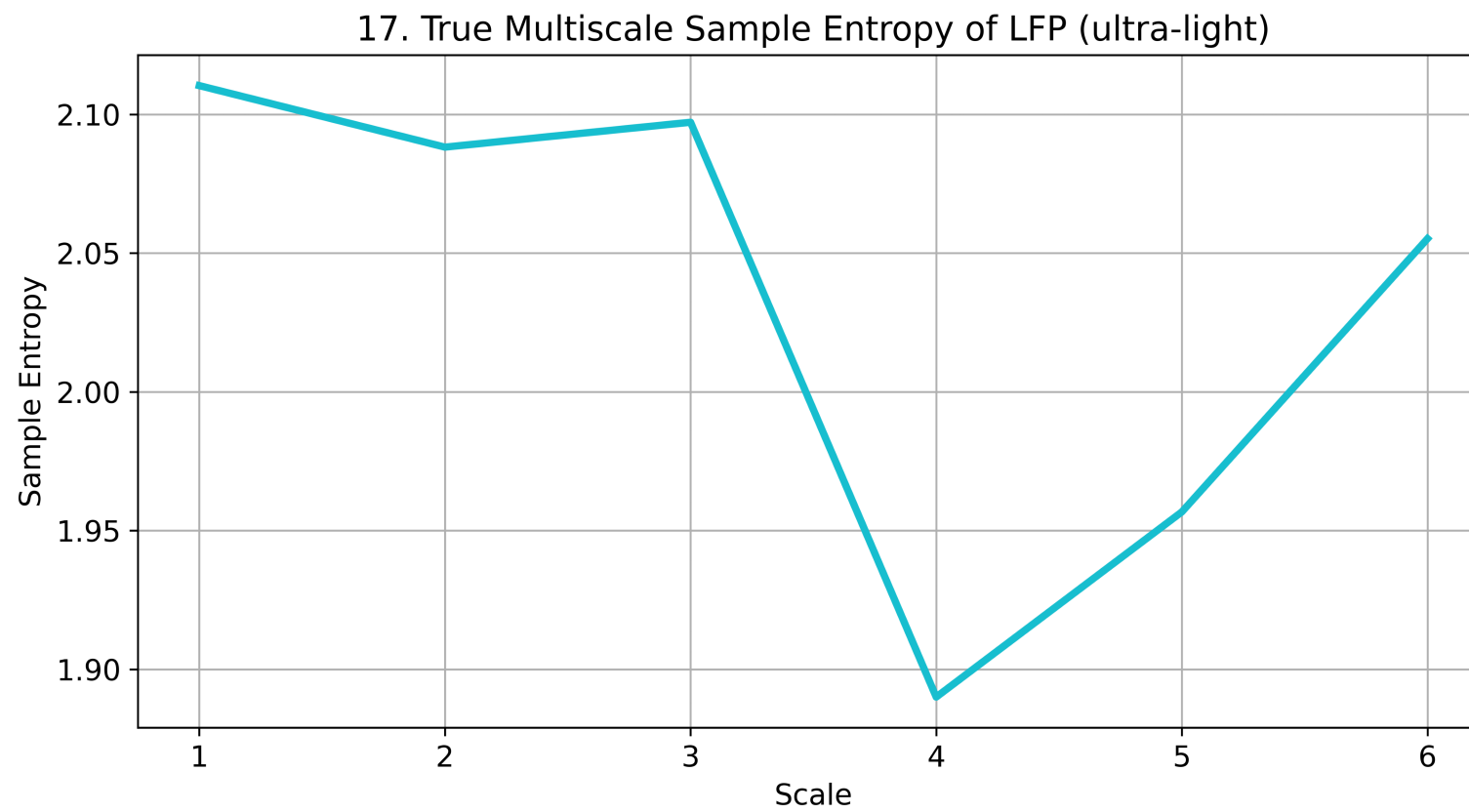
Buzsáki-style theta-43 Hz phase-amplitude coupling (Tort MI) ($t_{\text{peak}} \approx 69.789062$ s)



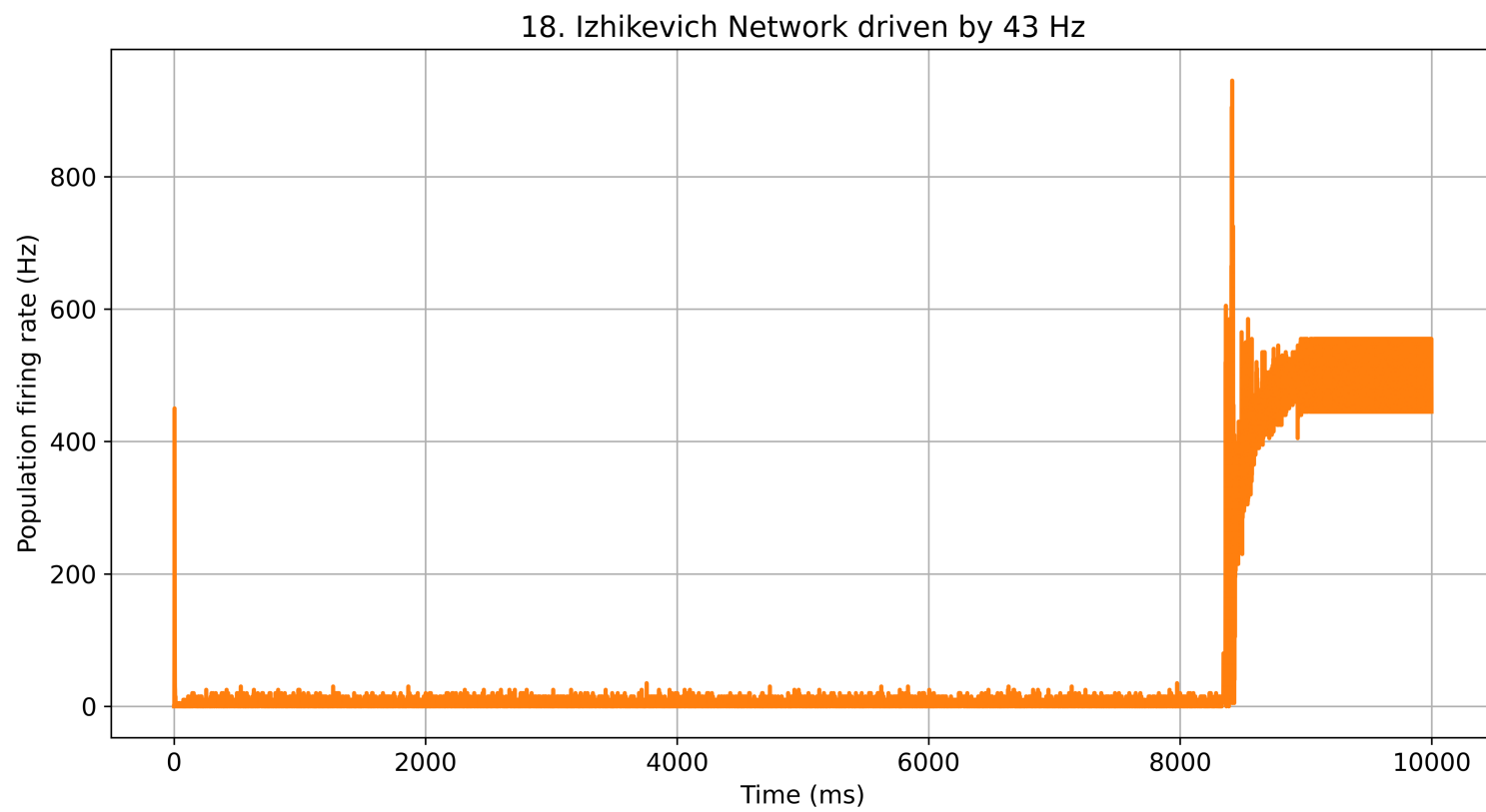
Gamma bursts locked to theta cycles (Buzsáki canonical) ($t_{\text{peak}} \approx 69.789062$ s)



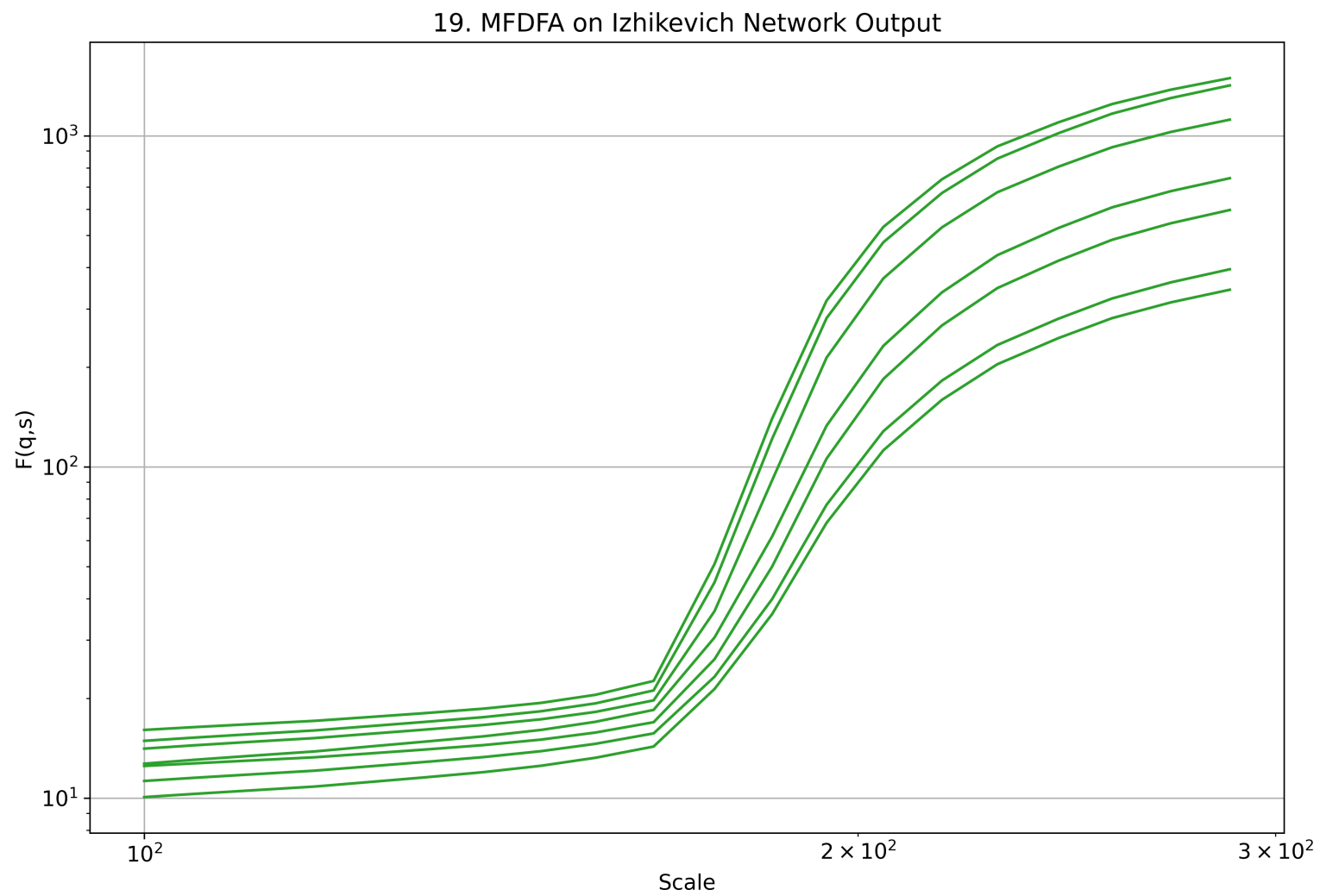
Welch LFP power spectrum showing 43 Hz peak ($t_{\text{peak}} \approx 69.789062$ s)



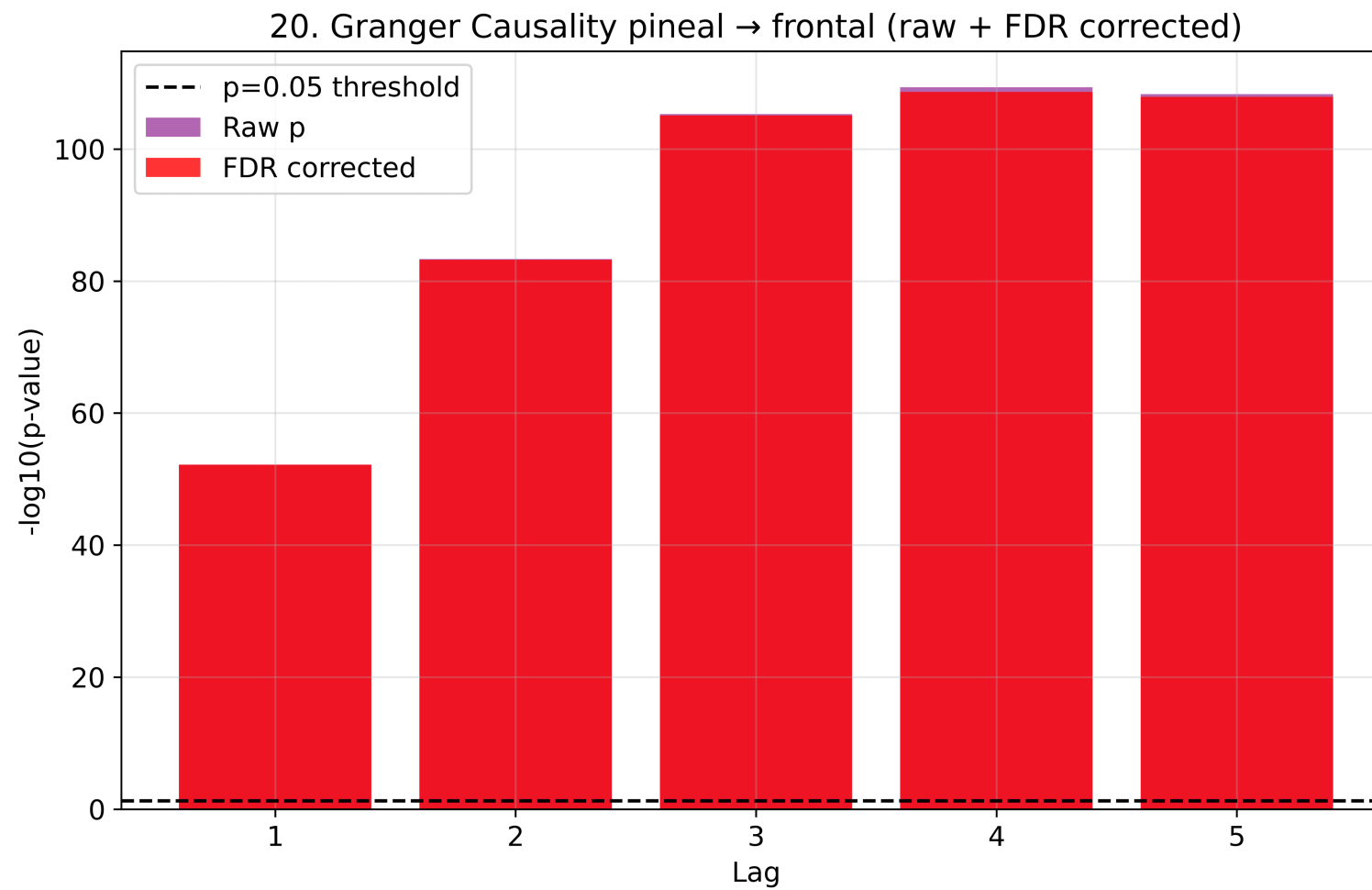
True multiscale sample entropy of LFP signal ($t_{\text{peak}} \approx 69.789062$ s)



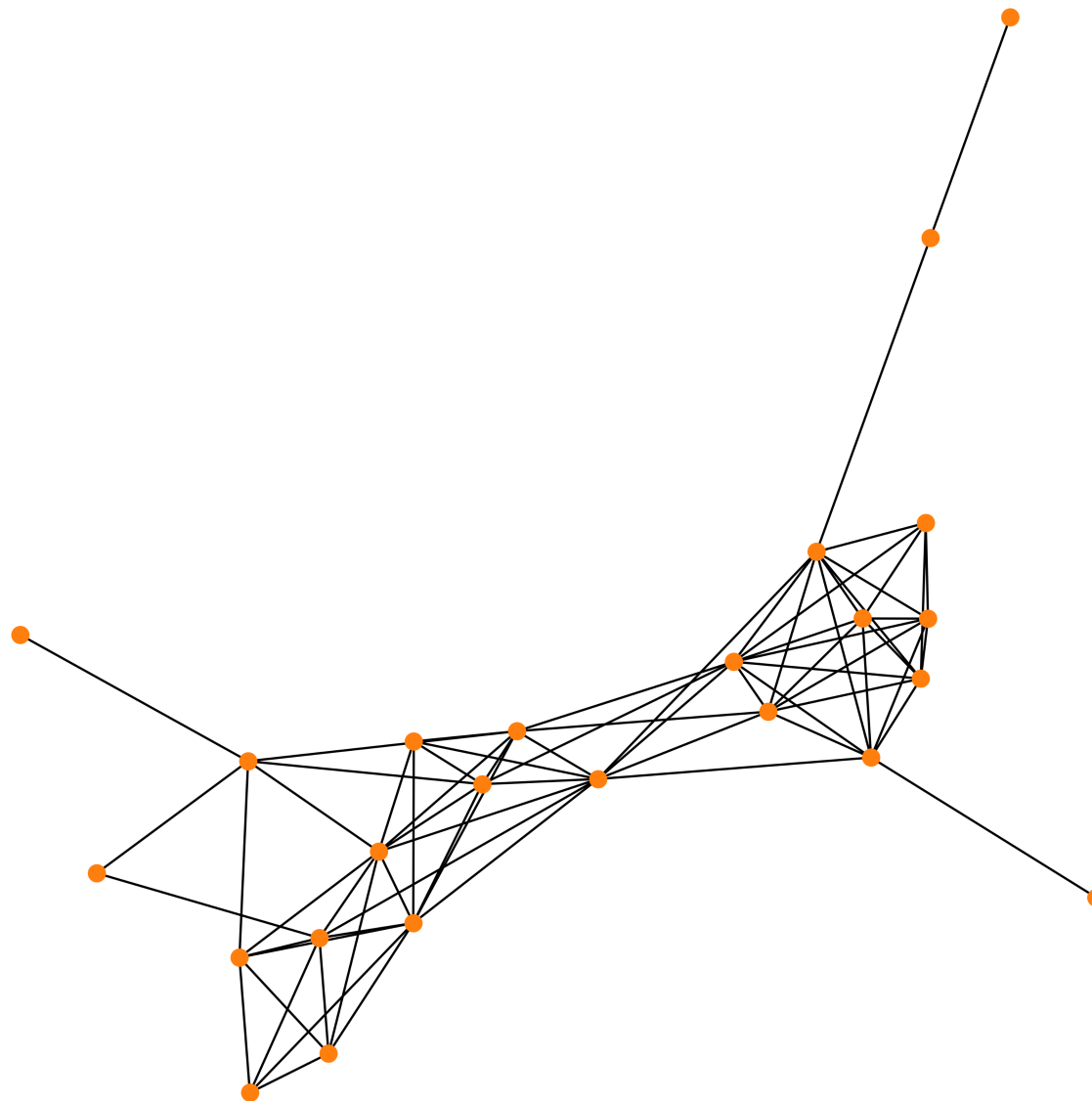
Izhikevich spiking network driven by observed 43 Hz gamma ($t_{\text{peak}} \approx 69.789062$ s)



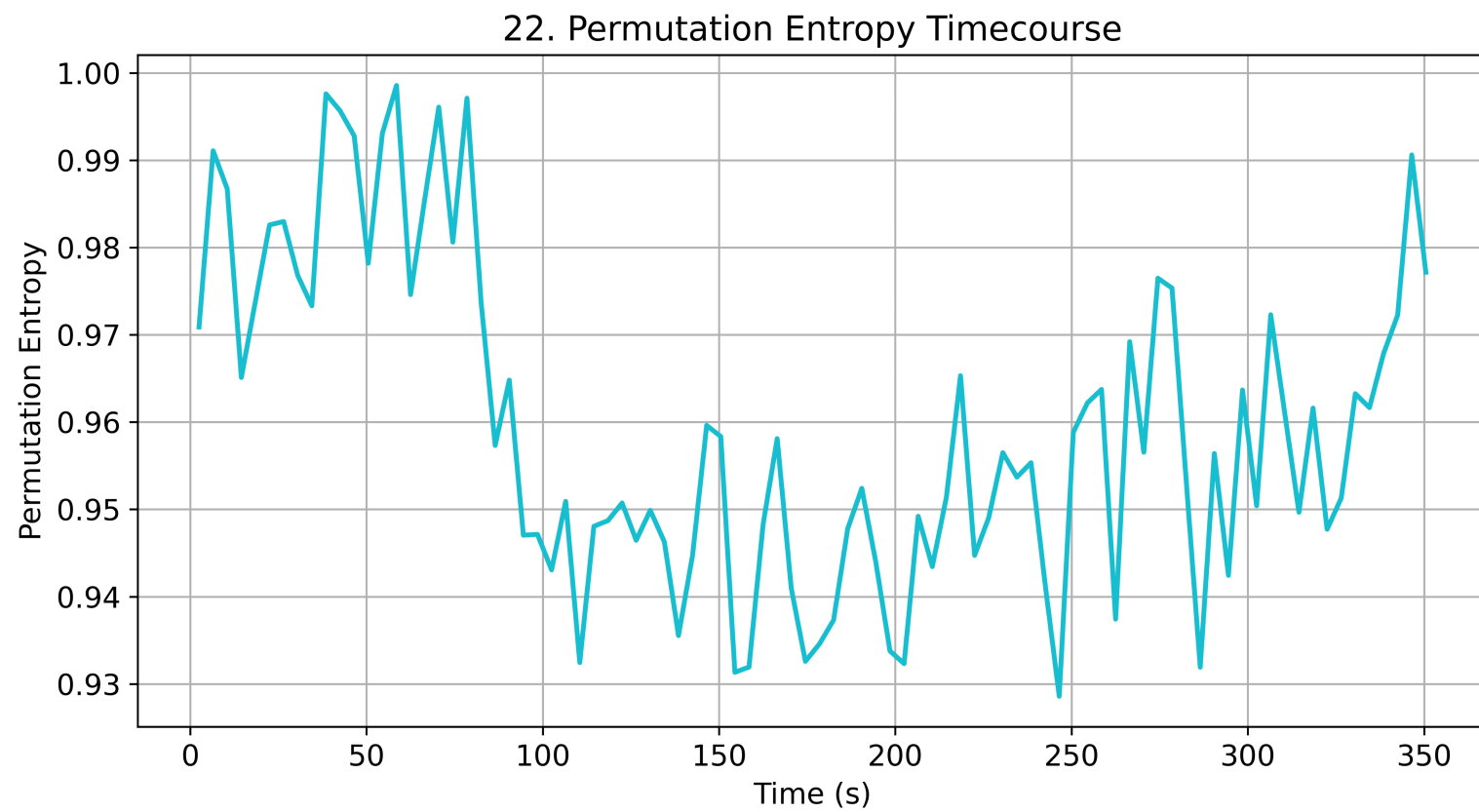
MF-DFA applied to Izhikevich network output ($t_{\text{peak}} \approx 69.789062$ s)



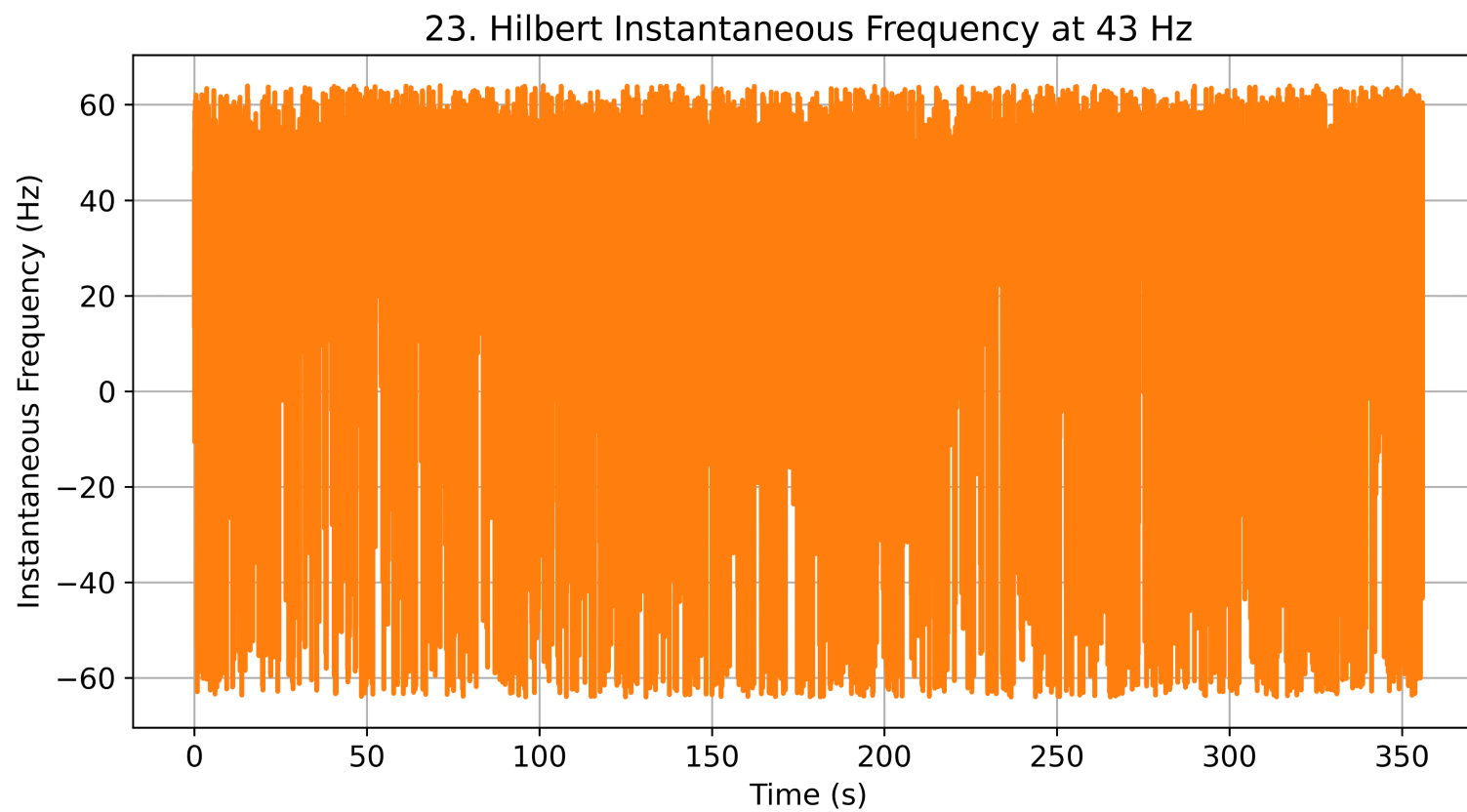
Granger causality pineal → frontal with FDR correction ($t_{\text{peak}} \approx 69.789062$ s)



Correlation-based EEG connectivity graph (no self-loops) ($t_{\text{peak}} \approx 69.789062$ s)



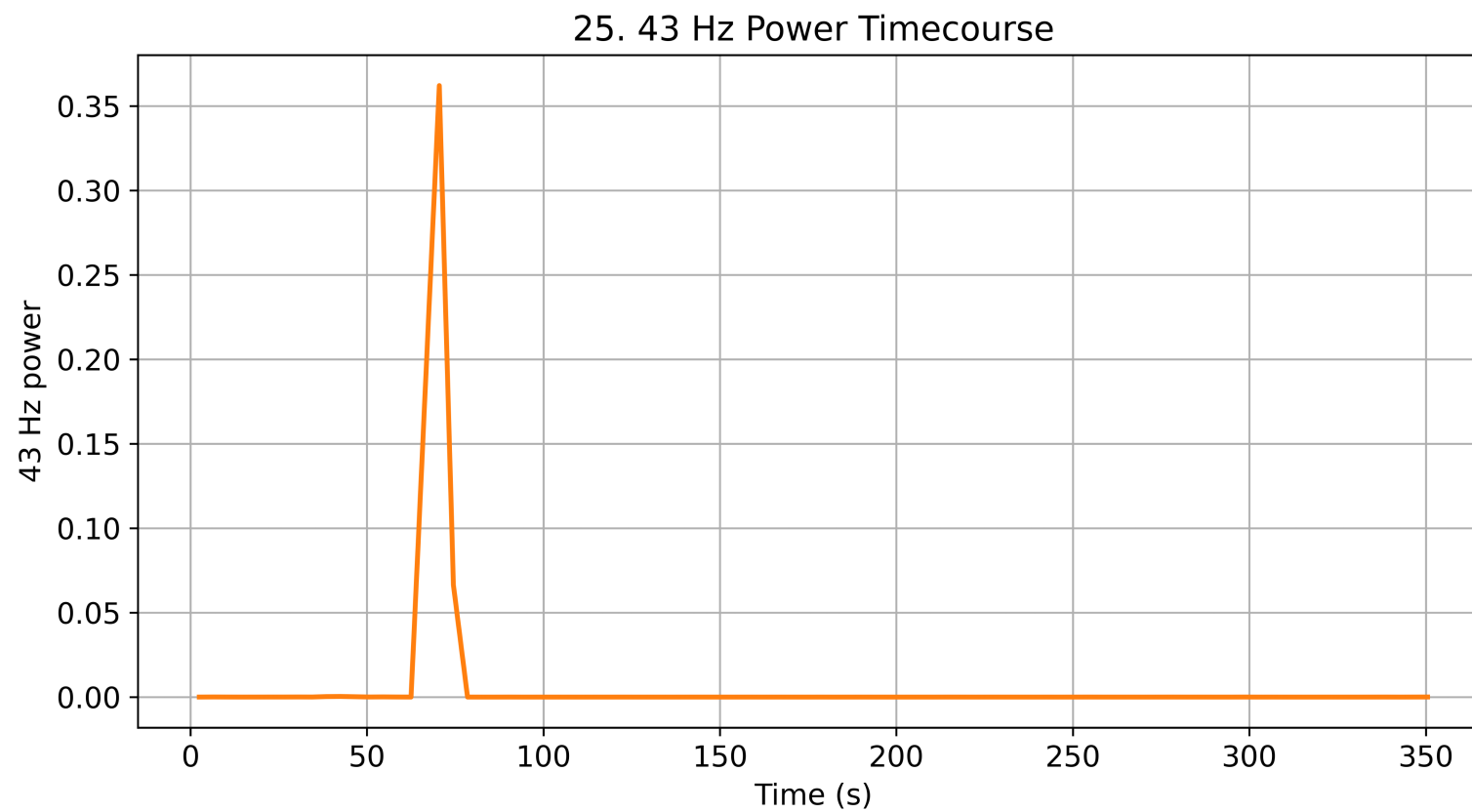
Permutation entropy timecourse ($t_{\text{peak}} \approx 69.789062$ s)



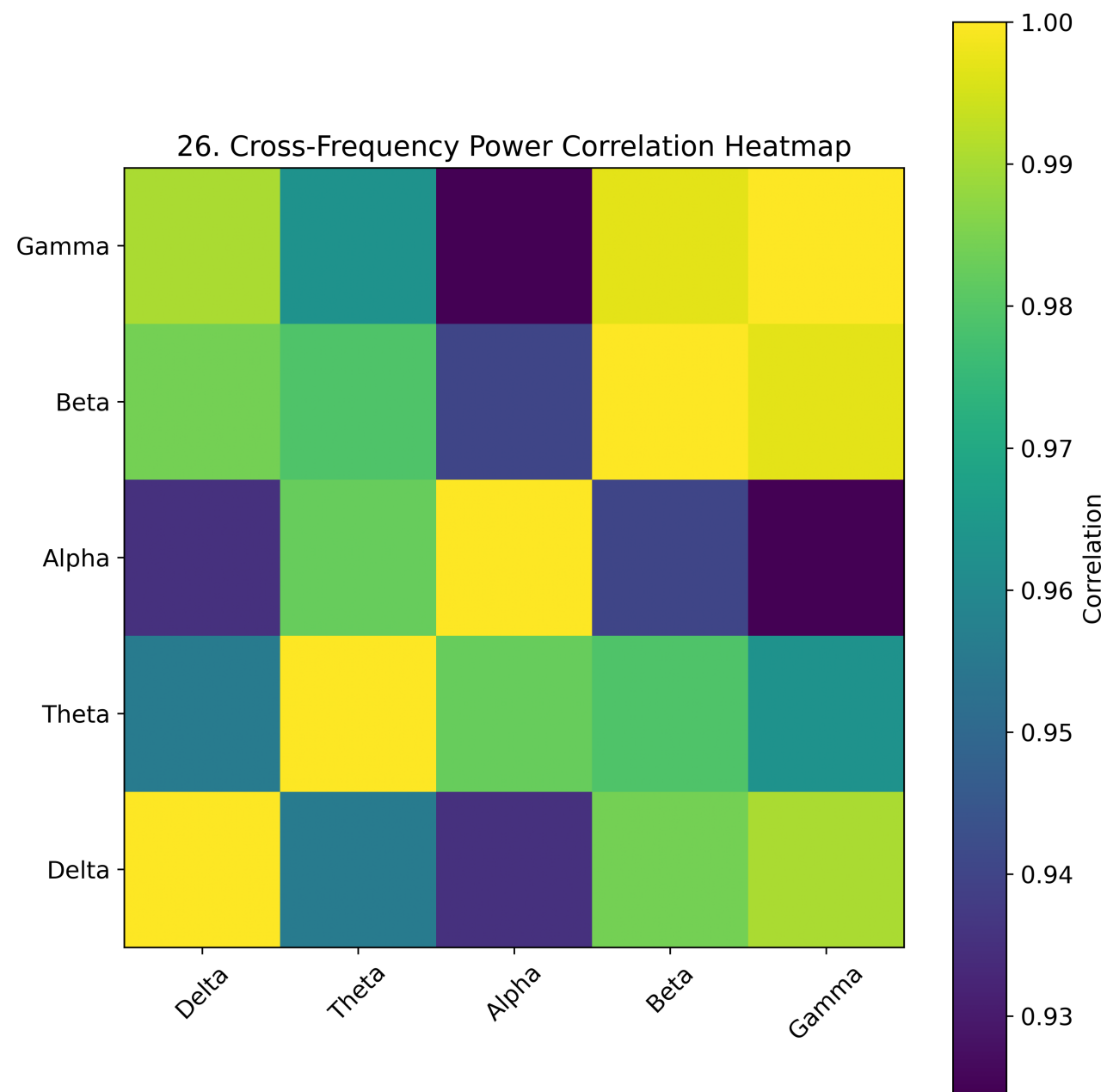
Instantaneous frequency of 43 Hz gamma component ($t_{\text{peak}} \approx 69.789062$ s)



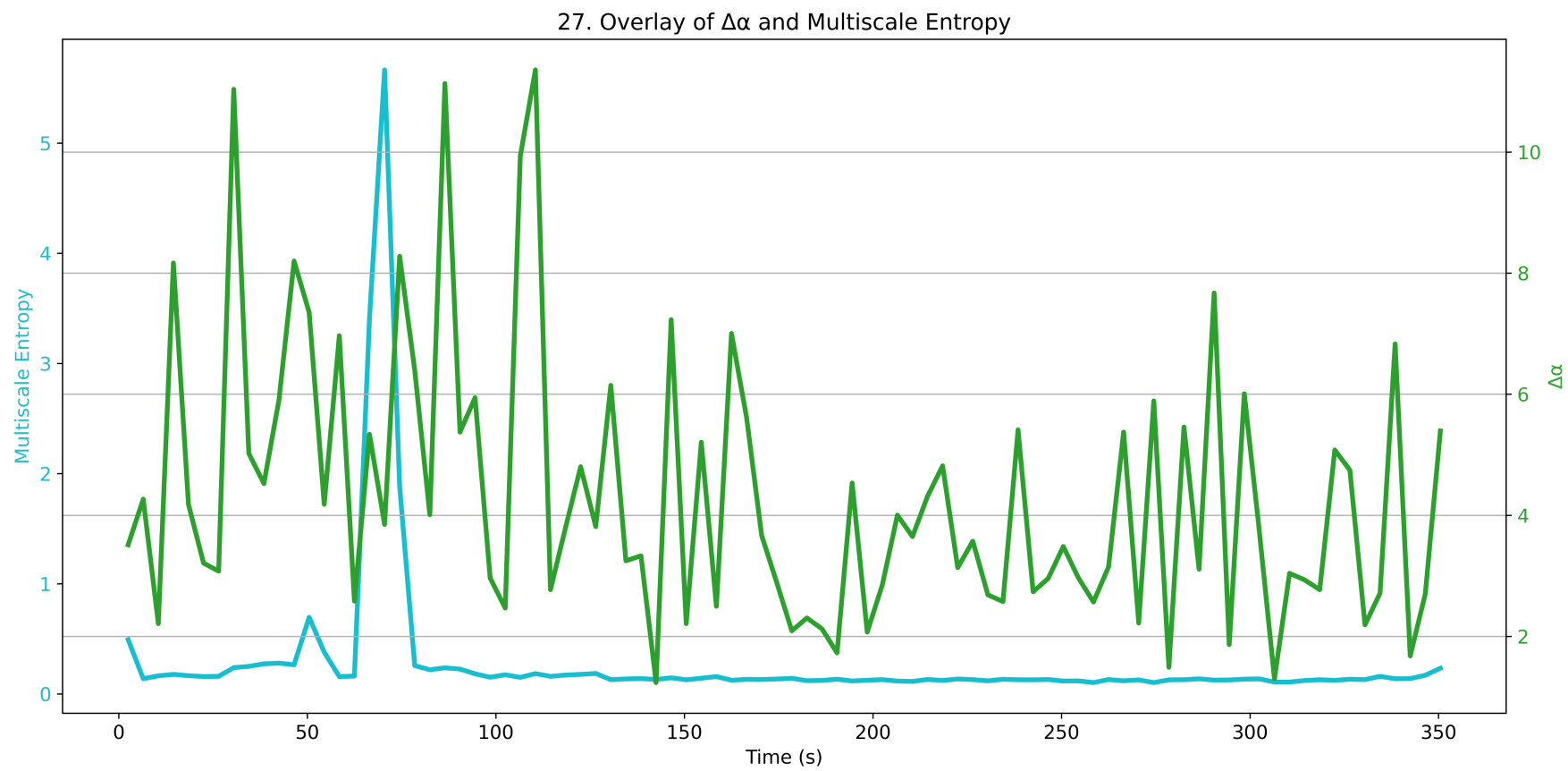
Theta (4-8 Hz) power timecourse ($t_{\text{peak}} \approx 69.789062$ s)



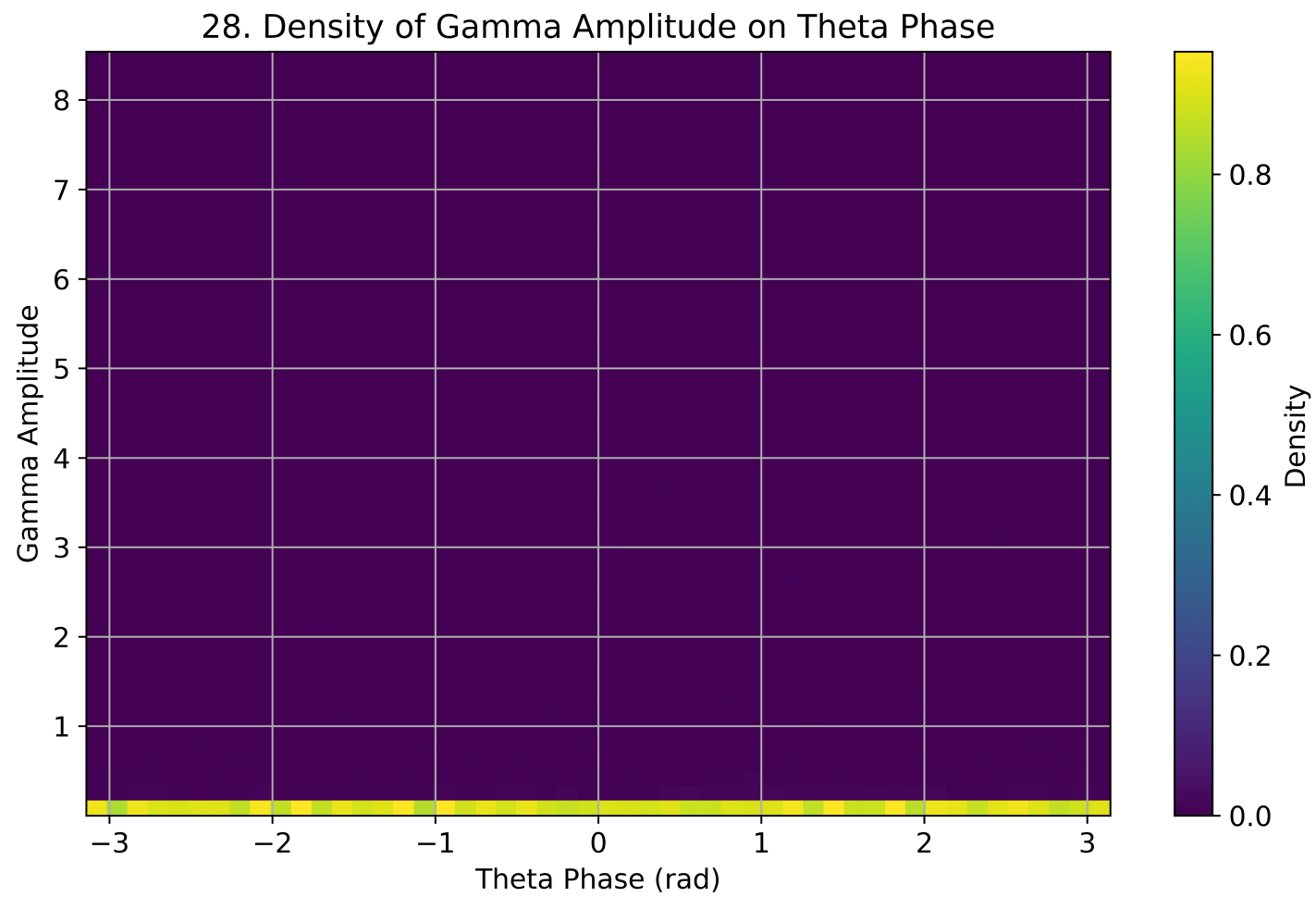
43 Hz gamma power timecourse ($t_{\text{peak}} \approx 69.789062$ s)



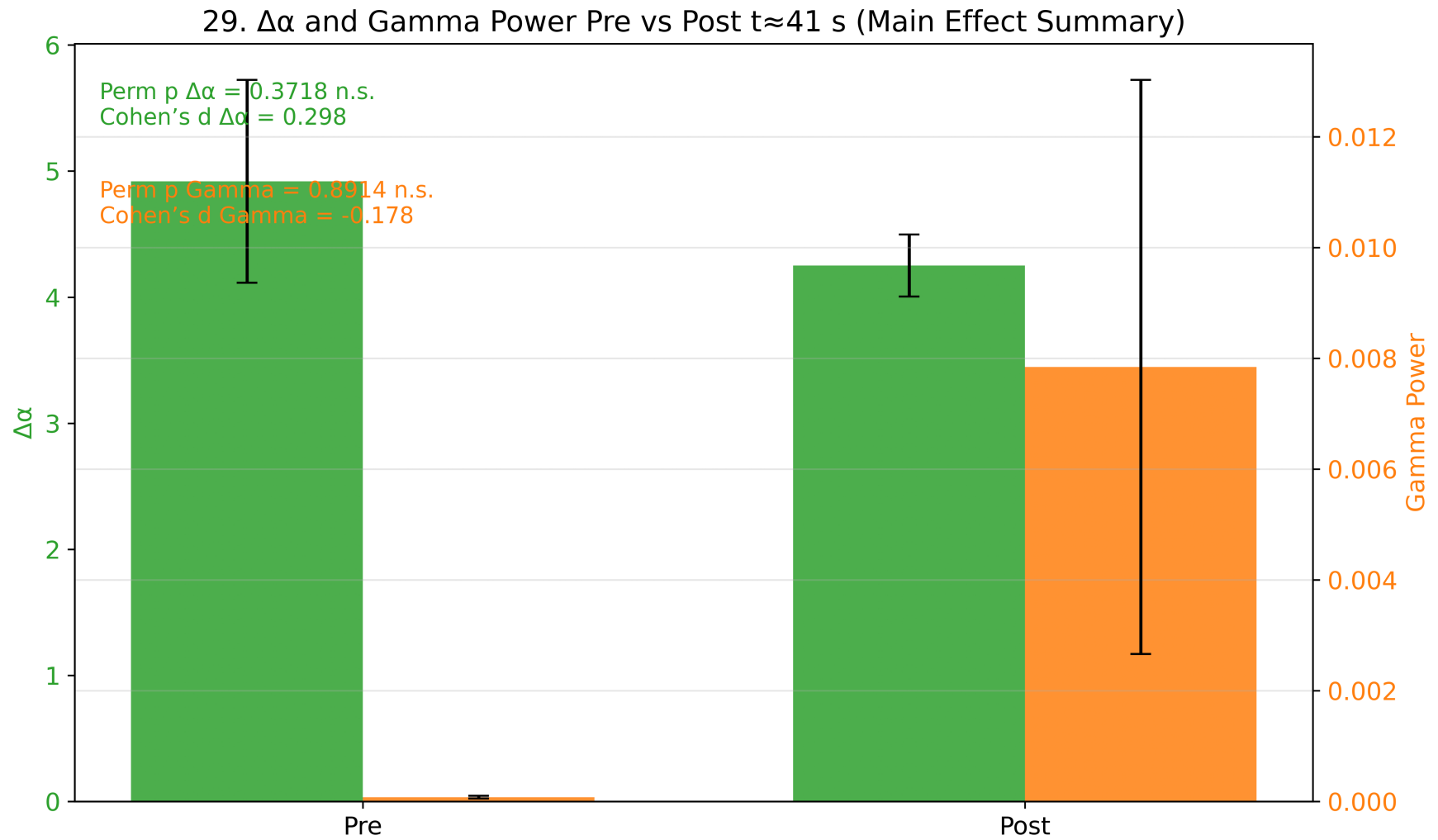
Cross-frequency power correlation heatmap ($t_{\text{peak}} \approx 69.789062$ s)



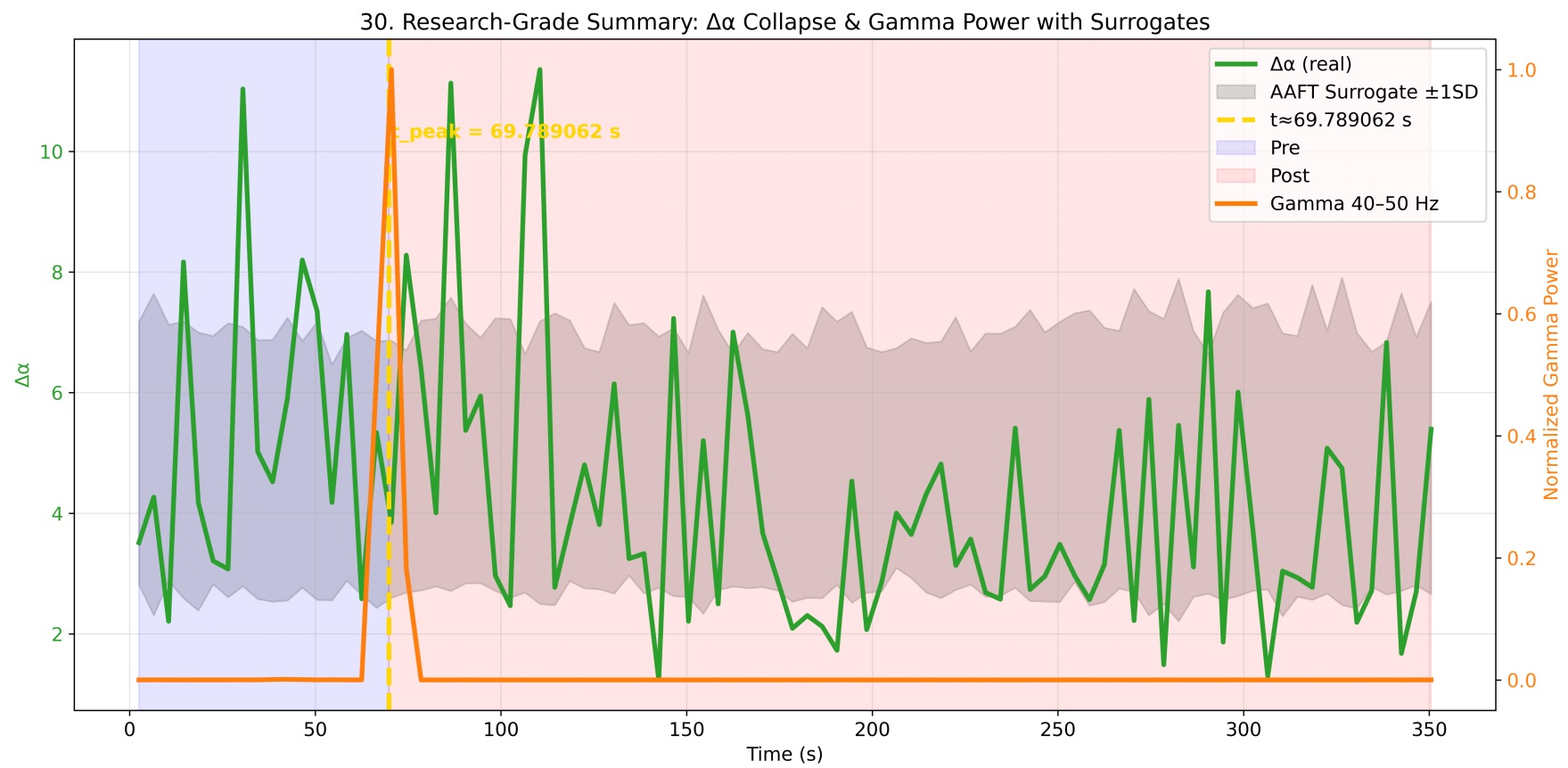
Overlay of $\Delta\alpha$ and multiscale entropy timecourses ($t_{\text{peak}} \approx 69.789062$ s)



2D density of gamma amplitude on theta phase ($t_{\text{peak}} \approx 69.789062$ s)



$\Delta\alpha$ and gamma power pre vs post $t \approx 69.789062$ s with Cohen's d and p-values



Research-grade summary: $\Delta\alpha$ collapse & gamma power with surrogate band and objective peak at 69.789062 s